

CI Outcome Prediction Project

Data Analysis Report, Baseline Replication & First Simple Models

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December 2025

1 Summary

1.1 Objective

This report documents our systematic analysis of the Swiss National CI Database for cochlear implant outcome prediction. The work has two goals:

1. **Replicate and validate** Bertschinger et al. (2024) — the most recent prediction model from USZ
2. **Extend the analysis** with uncertainty quantification, phenotype discovery, and cross-population generalization

1.2 Key Findings

- **Prediction ceiling is fundamental:** CV $R^2 \approx 0.10$ is robust to model choice, imputation method, feature engineering, outcome transformation, and temporal effects.
- **ML does not improve:** XGBoost, RF, LightGBM perform *worse* than linear regression due to overfitting.
- **Bertschinger’s features are near-optimal:** Lasso selection from 61 features retains 5/7 of her original predictors, improving R^2 by only +0.017.
- **Multi-Task Lasso achieves best R^2 :** Joint feature selection across all outcomes yields CV $R^2 = 0.18$ with 42 features—modest improvement but still clinically limited.
- **Simpson’s Paradox in pre-post correlation:** Overall $r = 0.16$ is *entirely* a between-group artifact. Within baseline strata, $r \approx 0$ (all $p > 0.05$). Pre-op WRS has **zero** predictive value for post-op WRS once baseline level is accounted for.
- **CI “Equalizer” effect:** All patients converge toward 60–70% post-op WRS regardless of starting point. Low baseline patients gain ~ 50 points; high baseline patients show minimal change or decline.
- **Decline is predictable:** 24% of patients show decreased WRS post-CI. These are high-baseline patients (mean pre = 69%). Logistic regression achieves CV AUC = 0.83 for predicting decline.
- **Latent factor models fail:** Pre and post scores do not load on a common factor—they measure fundamentally different systems (acoustic vs. electrical hearing).
- **Binary classification is more feasible:** Predicting success ($\geq 50\%$ WRS) achieves AUC = 0.70.
- **Uncertainty is large:** 90% prediction intervals span $\pm 39\%$ WRS — nearly the entire outcome range. No previous study reports this.
- **Phenotypes identified:** K-means clustering identifies 3 patient subgroups with significantly different outcomes ($p=0.0004$), but cluster membership does not improve prediction.

- **Adult models don't generalize:** Accuracy drops from 62% to 39% when applied to children.
- **Pediatric data gap:** Children <5 years (N=164, 70% of pediatric sample) cannot perform speech tests.

1.3 Interpretation

The prediction ceiling (CV $R^2 \approx 0.10$) and wide intervals ($\pm 39\%$) reflect **fundamental limitations of routine clinical data**, not model or methodology failure. We demonstrated this through extensive sensitivity analyses:

- 5 imputation methods $\rightarrow$ same R^2
- 4 model types (linear, Ridge, RF, GBM) $\rightarrow$ linear is best
- 6 outcome transformations $\rightarrow$ no improvement
- 61 features (49 available + 12 engineered) $\rightarrow$ overfitting
- All 226 variables with Multi-Task Lasso $\rightarrow$ CV $R^2 = 0.18$ (best, but still limited)
- Latent factor and multi-output models $\rightarrow$ no improvement
- Temporal detrending $\rightarrow$ no change
- Lasso feature selection $\rightarrow$ modest improvement (+0.017)

The Simpson's paradox finding explains *why* prediction is fundamentally limited: pre-operative tests measure **acoustic** hearing through a degraded peripheral system, while post-operative tests measure **electrical** hearing through a completely different transduction pathway. These are different systems—CI surgery “resets” the baseline-outcome relationship.

Key outcome determinants — auditory nerve health, cognitive function, rehabilitation quality — are not captured in current registries. Improving prediction requires biomarkers that assess neural capacity for electrical stimulation (eCAP, imaging-derived neural density, cognitive tests), not acoustic hearing status.

Novel contributions:

1. First study to identify Simpson's paradox in CI outcome prediction
2. First study to report prediction intervals for CI outcomes
3. Clinical counseling guide based on baseline-stratified outcomes

1.4 Methods Overview

Analysis	Method	Status
Bertschinger replication	Linear regression, ABESS, missForest	Complete
ML benchmark	Ridge, RF, XGBoost, LightGBM	Complete
Extended features	49 available + 12 engineered, Lasso	Complete
Extended predictors	All 226 variables, Multi-Task Lasso, PLS	Complete
Pre-post correlation	Simpson's paradox, stratified analysis	Complete
Latent factor models	Multi-output, factor analysis, interactions	Complete
Temporal validation	Train ≤ 2021 , test ≥ 2022	Complete
Binary classification	Logistic regression, AUC analysis	Complete
Sensitivity analyses	Imputation, transformations, temporal	Complete
Uncertainty quantification	Split conformal prediction, CQR	Complete
Phenotype discovery (RQ1)	K-means, Hierarchical, GMM clustering	Complete
Cross-population	Adults $\rightarrow$ Children transfer	Complete

1.5 Document Structure

This report is organized as follows:

- **Section 2 (Dataset Overview):** Data source, filtering pipeline, demographics, outcome distribution
- **Section 3 (Prediction Analysis — RQ2):** Bertschinger replication, ML benchmark, feature engineering, extended predictor analysis (all 226 variables), temporal validation, binary classification, prediction ceiling summary
- **Section 4 (Pre-Post Correlation Structure):** Simpson’s paradox, the “Equalizer” effect, decline analysis, latent factor models, clinical implications
- **Section 5 (Uncertainty Quantification):** Conformal prediction intervals, CQR, coverage analysis
- **Section 6 (Sensitivity Analyses):** Imputation methods, coefficient stability, outcome transformations, temporal stability, model complexity
- **Section 7 (Phenotype Discovery — RQ1):** Clustering analysis, patient subgroups, implications for prediction
- **Section 8 (Adult vs. Pediatric):** Cross-population generalization, outcome engineering
- **Section 9 (Pediatric Data Gap):** Missing outcomes for children <5, data request
- **Supplementary Information:** Prediction details, imputation comparison, clustering algorithm comparison, data quality diagnostics

2 Dataset Overview

2.1 Data Source

Two datasets were analyzed:

- **Adults-only:** CI_Outcome_Prediction_20241113.xlsx (N=479)
- **Combined:** CI_Outcome_Prediction_20251126_incl_children.xlsx (N=771)

Step	N	%	Excluded
Full dataset (adults + children)	771	–	–
Adults only (≥ 18 years)	479	62%	292 children
With post-op WRS (Mono1_post)	385	80%	94 missing outcome
After SSD exclusion	326	85%	59 SSD/unknown
<i>Bertschinger et al. (2024)</i>	360	–	–

Table 1: Dataset filtering pipeline. Percentages show retention rate from previous step.

2.2 Data Pipeline Summary

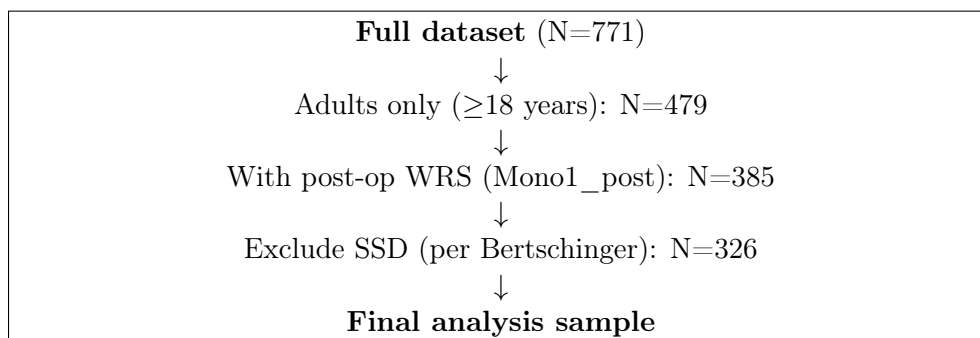


Figure 1: Data filtering pipeline showing patient flow from raw data to analysis sample.

2.3 Variable Overview

The Swiss National CI Database contains 226 variables per patient, organized into the following categories:

Category	N vars	Availability	Examples
Demographics	3	100%	Age, sex, CI date
Pure-tone audiometry	16	98%	PTA at 125–8000 Hz, both ears
Free-field audiometry	14	65–85%	Aided thresholds at 250–4000 Hz
Speech tests (pre-op)	6	58–62%	Mono, FM, Num, OLSA, Consonants
Speech tests (post-op)	6	80–88%	Same tests, post-implant
Communication	15	65–99%	Articulation, language, phone use
Evaluation/History	26	6–100%	Etiology, onset, HL duration
Imaging (CT)	11	65–85%	Anatomical abnormalities
Imaging (MRI)	6	80–98%	Cochlear/nerve status
Socioeconomic	9	0–97%	Education, insurance, employment
Surgical/Postoperative	14	45–97%	Device, complications
Other	108	Variable	IDs, dates, derived variables

Table 2: Variable categories in the Swiss National CI Database (N=226 total variables).

2.3.1 Speech Outcome Measures

Six speech tests were administered pre- and post-operatively:

Test	Description	N (pre)	N (post)	Range
Mono1/Mono2	Freiburg monosyllables (WRS)	280 (58%)	385 (80%)	0–100%
FM	Freiburg multisyllables	298 (62%)	405 (85%)	0–100%
Num	Freiburg numbers	290 (61%)	403 (84%)	0–100%
V08	OLSA sentence test	287 (60%)	419 (87%)	dB SNR
C12	Consonant confusion test	277 (58%)	410 (86%)	0–100%

Table 3: Speech outcome measures. Monosyllables (Mono1) is the primary outcome following Bertschinger et al. (2024). Pre-operative tests are available for only ~60% of patients.

2.3.2 Communication and Functional Variables

Variable	Categories	N	Availability
COM_ARTICULATION	Normal / Slight dist. / Strong dist.	462	96.5%
COM_MAIN_LANGUAGE	German, Swiss-German, other (21 total)	472	98.5%
COM_PHONE_USE	Free use / Known persons / Simple msg / Never	421	87.9%
COM_LIP_READING	Yes / No	309	64.5%
COM_Profession	Communication demands at work	286	59.7%

Table 4: Key communication variables used in prediction models.

2.3.3 Imaging Variables

Pre-operative imaging provides information on cochlear and neural anatomy:

Variable	Description	N	Abnormal
<i>CT findings (N=409 performed, 85%)</i>			
CT_Cochl_malform	Cochlear malformation	314	1.3%
CT_Laby_ossi	Labyrinthine ossification	312	4.2%
CT_Mastoid_middle_ear_abnorm	Middle ear abnormality	317	8.5%
<i>MRI findings (N=469 performed, 98%)</i>			
MRI_Cochl_fibr	Cochlear fibrosis	383	4.4%
MRI_Nerve_Tumor	Auditory nerve tumor	381	3.9%
MRI_Cochl_Nerve_absence	Cochlear nerve absence	382	0.0%

Table 5: Imaging findings. MRI was performed in 98% of patients; CT in 85%. Abnormalities are relatively rare.

2.3.4 Socioeconomic Variables

Variable	Description	N	Availability
SES_EDUCATION_LEVEL	Highest education attained	431	90.0%
SES_HEALTH_INSURANCE	Insurance type	466	97.3%
SES_WORKING	Currently employed	391	81.6%
SES_LIVING_SITUATION	Living arrangement	275	57.4%
SES_EMPLOYMENT_TYPE	Full-time / Part-time / Other	228	47.6%

Table 6: Socioeconomic variables. Education level has good availability (90%) and was selected by Lasso as a predictor.

2.3.5 Device Information

Variable	Description	N	Availability
Modell_CI	CI device model (numeric code)	479	100%
Datum_ErstesCI	Date of first CI	479	100%
Datum_KontralatCI	Date of contralateral CI (if bilateral)	75	15.7%

Table 7: Device and surgical timing variables. 15.7% of patients received bilateral implants. Device models are stored as numeric codes.

2.3.6 Variables Not Available

Several variables that might improve prediction are **not captured** in the database:

- **Auditory nerve health:** Electrically-evoked compound action potentials (eCAP), neural response telemetry
- **Cognitive function:** Working memory, processing speed, executive function
- **Rehabilitation intensity:** Hours of auditory training, speech therapy frequency
- **Device fitting parameters:** Mapping levels, compression ratios, active electrodes
- **Daily device use:** Hours/day, listening environments
- **Quality of life:** Subjective benefit, satisfaction scores (SSQ, NCIQ)

These unmeasured factors likely explain the prediction ceiling ($CV R^2 \approx 0.10$) observed in Section 3.

2.4 Demographics

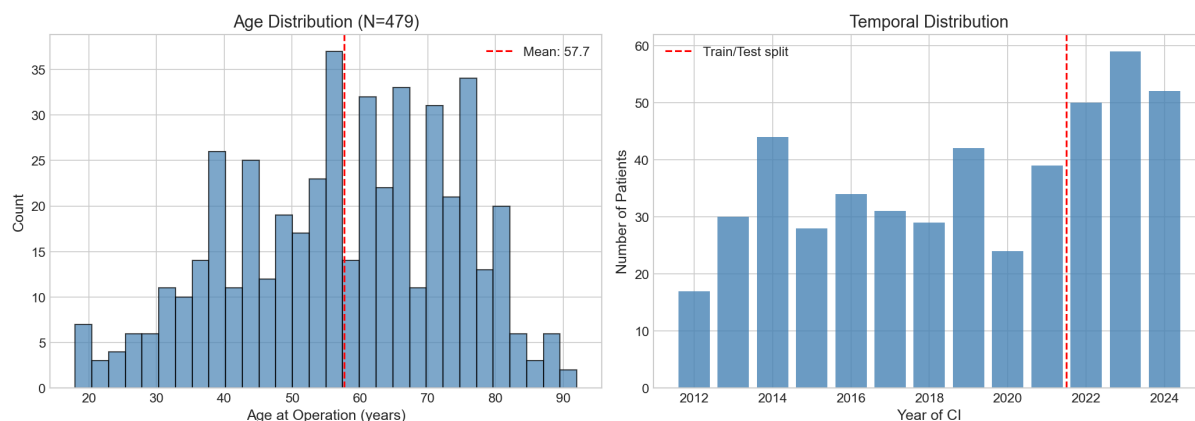


Figure 2: Left: Age distribution at implantation (mean 57.7 years). Right: Temporal distribution of CI procedures showing COVID-19 dip in 2020 and growth from 2022–2024. Red dashed line indicates proposed train/test split at 2021.

Key demographic characteristics:

- **Age at implant:** Mean 57.7 years (range 18–93)
- **Sex:** 50.7% female, 49.3% male
- **Duration of hearing loss:** Mean 23.1 years (Figure 3)
- **Time period:** 2012–2024
- **Pre-op WRS:** Mean 41.6% (N=280, 58.5% available)
- **Post-op WRS:** Mean 61.1% (SD 26.2%, N=385)
- **Binary outcome:** 72% success ($\geq 50\%$ WRS), 28% failure

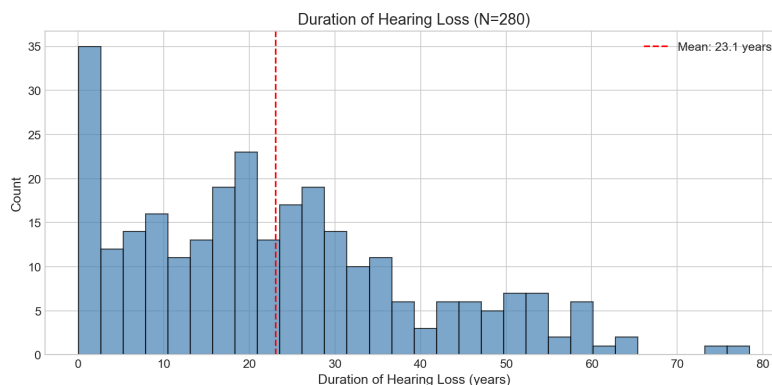


Figure 3: Distribution of hearing loss duration prior to implantation. Note bimodal pattern with spike at 0–5 years (sudden/progressive onset) and broader distribution reflecting congenital or early-onset cases.

2.5 Categorical Predictors

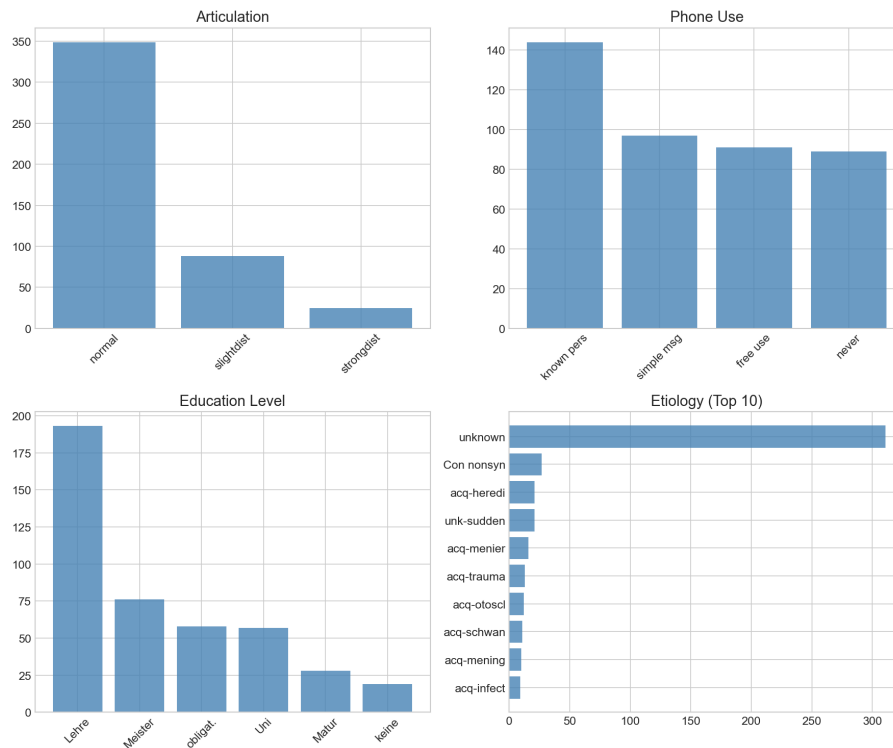


Figure 4: Distribution of key categorical predictors: articulation quality, phone use capability, education level, and etiology (top 10 categories). Note high proportion of unknown etiology.

2.6 Hearing Loss Classification

Following Bertschinger's methodology (Van De Heyning et al. 2017), patients were classified by pure-tone average (PTA):

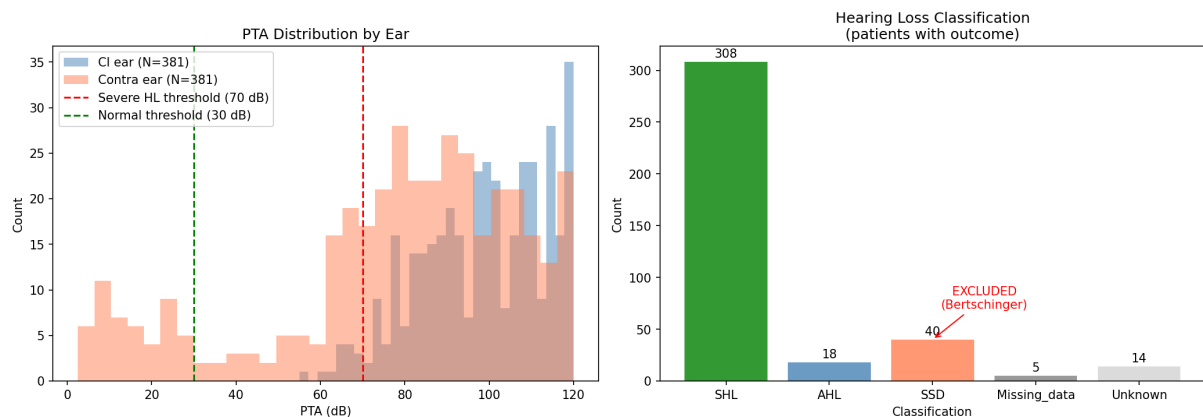


Figure 5: Left: PTA distribution by ear showing CI ear (severe-profound) vs. contralateral ear (variable). Right: Hearing loss classification with SSD patients excluded per Bertschinger methodology. PTA values capped at 120 dB.

Classification	Criteria	N	Included?
SHL (Symmetric)	PTA CI ≥ 70 dB, PTA Co > 55 dB	308	Yes
AHL (Asymmetric)	PTA CI ≥ 70 dB, PTA Co 30–55 dB	18	Yes
SSD (Single-Sided)	PTA CI ≥ 70 dB, PTA Co < 30 dB	40	Excluded
Missing/Unknown	Incomplete PTA data	19	Excluded

Table 8: Hearing loss classification (among 385 patients with outcome). SSD excluded to match Bertschinger.

2.7 Outcome Distribution

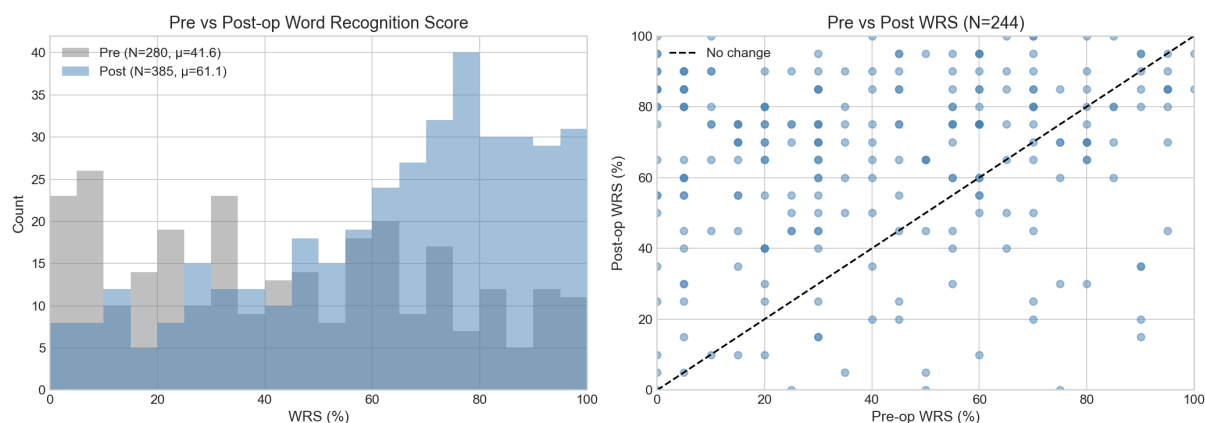


Figure 6: Left: Pre-operative (gray) vs. post-operative (blue) WRS distributions showing clear shift toward higher scores. Right: Individual patient trajectories; points above diagonal indicate improvement. Note 21.8% of patients show decreased WRS post-CI.

2.8 Predictor Availability

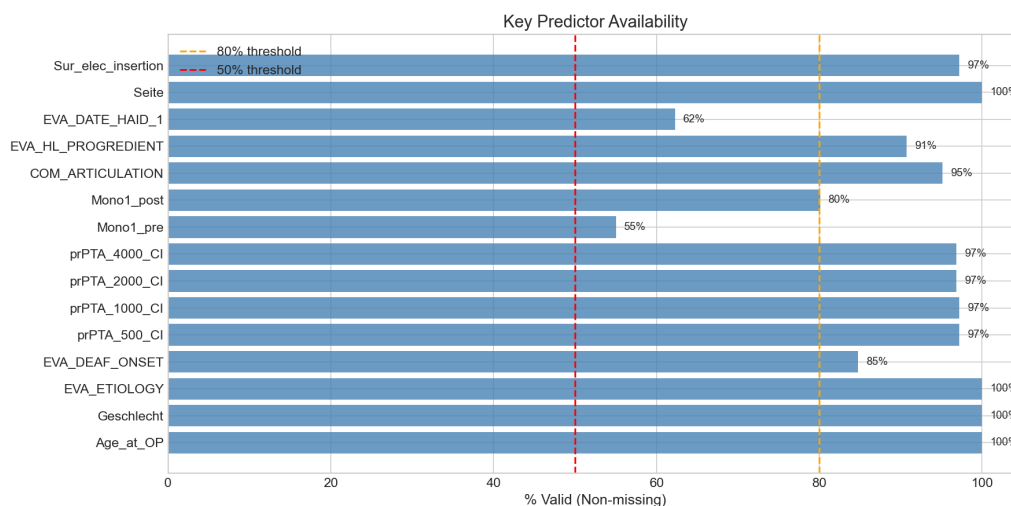


Figure 7: Key predictor availability in the adult dataset (N=479). Core variables (age, sex, etiology, PTA) have $> 95\%$ availability. Pre-operative WRS (Mono1_pre) is available for only 55% of patients, which is why it was not included as a predictor despite its potential predictive value. Audiometric values were capped at 120 dB.

3 Prediction Analysis (RQ2)

RQ2 asks: Can we predict cochlear implant outcomes from pre-operative clinical data?

We addressed this through a systematic series of analyses: (1) replicating the current benchmark model, (2) testing machine learning alternatives, (3) expanding the feature set, (4) validating on future patients, and (5) exploring binary classification. The consistent finding across all approaches is a prediction ceiling of $CV\ R^2 \approx 0.10$.

We approached prediction through a systematic series of questions, each designed to isolate potential sources of improvement:

- Can we replicate the benchmark? (Section 3.1) — Establishes baseline performance
- Do more complex models help? (Section 3.2) — Tests if non-linear relationships exist
- Do more features help? (Section 3.3) — Tests if information is missing from the model
- Does the model generalize temporally? (Section 3.4) — Tests if performance is stable over time
- Is binary prediction easier? (Section 3.5) — Tests if the outcome formulation is the problem

Each negative finding narrows the explanation: the prediction ceiling reflects fundamental data limitations—unmeasured factors like neural health, cognitive function, and rehabilitation quality—rather than methodological choices.

3.1 Baseline: Bertschinger Replication

Bertschinger et al. (2024) developed the most recent prediction model for CI outcomes using the Swiss National CI Database. We replicated her methodology to establish a baseline.

3.1.1 Methodology Comparison

Step	Bertschinger	Our Implementation
Population	Adults ≥ 18 , first CI, 2012+	Same
Exclude SSD	Yes	Yes
Audiometry capping	120 dB	120 dB
Imputation	missForest	IterativeImputer (RF)
Feature selection	ABESS (24 $\rightarrow$ 7)	ABESS (24 $\rightarrow$ 7)
Model	Linear regression	Linear regression
Outcome transformation	Z-score	Z-score

Table 9: Methodological comparison showing close alignment with Bertschinger et al. (2024).

3.1.2 Feature Selection: ABESS Reproducibility

We ran ABESS independently on our data (N=326) with 24 candidate predictors. The algorithm selected 7 features, with 6 of 7 matching Bertschinger’s selection:

Feature	Our ABESS	Bertschinger	Match?
Articulation (moderate)	✓	✓	✓
Articulation (poor)	✓	✓	✓
German speaker	✓	✓	✓
Etiology: Schwannoma	✓	✓	✓
Duration of HL (years)	✓	✓	✓
Free-field CI (dB)	✓	✓	✓
Education level	✓	–	Ours only
Phone use (limited)	–	✓	Hers only

Table 10: ABESS feature selection comparison. 6 of 7 features are identical, demonstrating reproducibility.

3.1.3 Performance Results

Model	N	Features	R ² (train)	CV R ²	MAE
Bertschinger’s exact 7	326	7	0.175	0.102 ± 0.10	19.4%
Our ABESS selection	326	7	0.176	0.101 ± 0.10	19.4%
<i>Bertschinger (train)</i>	<i>360</i>	<i>7</i>	<i>0.274</i>	<i>N/A</i>	<i>13.3%</i>
<i>Bertschinger (validation)</i>	<i>16</i>	<i>7</i>	<i>–</i>	<i>–</i>	<i>17.2%</i>

Table 11: Performance comparison. Our CV R² $\approx$ 0.10 represents true generalization performance. The gap with Bertschinger’s reported R² (0.27) reflects training vs. cross-validated comparison.

3.1.4 Coefficient Comparison

All seven coefficient signs matched Bertschinger’s original model, though magnitudes were consistently smaller in our replication. This confirms that the predictor–outcome relationships are robust across samples.

Baseline finding: CV R² $\approx$ 0.10 likely represents true generalization performance. This aligns with Bertschinger’s validation MAE (17.2%) $\approx$ our CV MAE ($\sim$ 19%) and literature benchmarks (Shafieibavani et al. 2021: R² $\approx$ 0.12).

3.2 Machine Learning Benchmark

We tested whether non-linear machine learning models could capture additional signal beyond linear regression.

3.2.1 Models Evaluated

- **Ridge Regression** (linear baseline with L2 regularization)
- **Random Forest** (ensemble of decision trees)
- **XGBoost** (gradient boosted trees)
- **LightGBM** (gradient boosted trees, optimized)

All models used 17 features (Bertschinger’s features plus additional clinical variables). Hyperparameters were tuned via nested cross-validation.

3.2.2 Cross-Validation Results

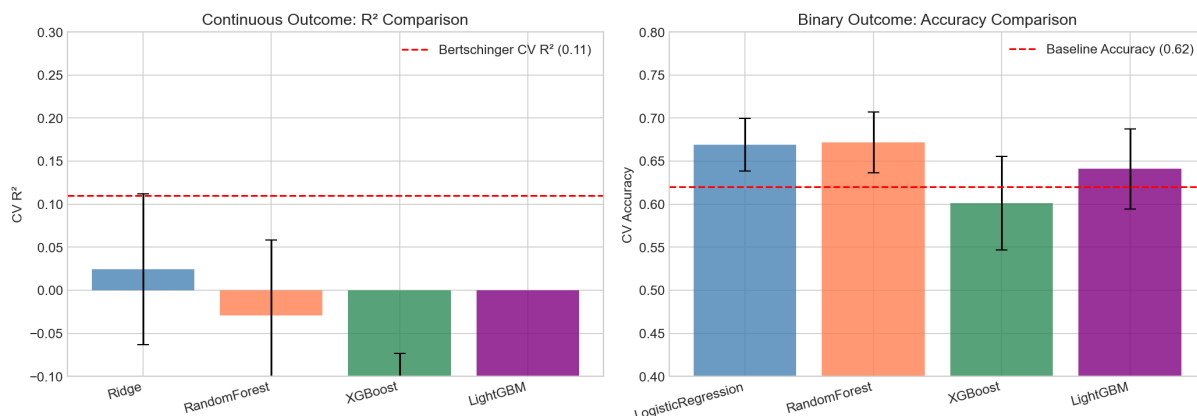


Figure 8: Cross-validation comparison. Left: Continuous outcome (R^2). Right: Binary outcome (Accuracy). Red dashed lines show baseline performance. ML models do not improve over linear regression.

Model	CV R^2	CV MAE	CV Accuracy	CV AUC
Ridge (linear)	0.024 $\pm$ 0.09	20.9	0.669 $\pm$ 0.03	0.653
Random Forest	-0.029 $\pm$ 0.09	21.4	0.672 $\pm$ 0.04	0.625
XGBoost	-0.164 $\pm$ 0.09	22.6	0.601 $\pm$ 0.06	0.605
LightGBM	-0.230 $\pm$ 0.13	22.9	0.641 $\pm$ 0.05	0.578

Table 12: Cross-validation results. Negative R^2 indicates models perform worse than predicting the mean. Linear model (Ridge) achieves best continuous prediction.

3.2.3 Feature Importance

Feature importance analysis from XGBoost confirms that top predictors align with Bertschinger's selection: articulation, German speaker, free-field thresholds, and duration of hearing loss.

ML finding: Machine learning does not improve over linear regression. XGBoost and LightGBM perform *worse* (negative R^2) due to overfitting on the small sample ($N=326$). The predictor–outcome relationships appear genuinely linear.

3.3 Feature Engineering & Lasso Selection

We tested whether additional features could improve prediction beyond Bertschinger's 7 predictors.

3.3.1 Feature Engineering

Part A: All available features. We extracted 49 features from the dataset, including individual audiometric frequencies, imaging findings, surgical variables, and socioeconomic factors.

Part B: Engineered features. We created 12 derived features including PTA asymmetry, hearing reserve, age $\times$ duration interaction, audiometric slope, and communication composites.

Part C: Feature selection. We applied Lasso regression with cross-validated regularization to select informative features from the combined set.

3.3.2 Results

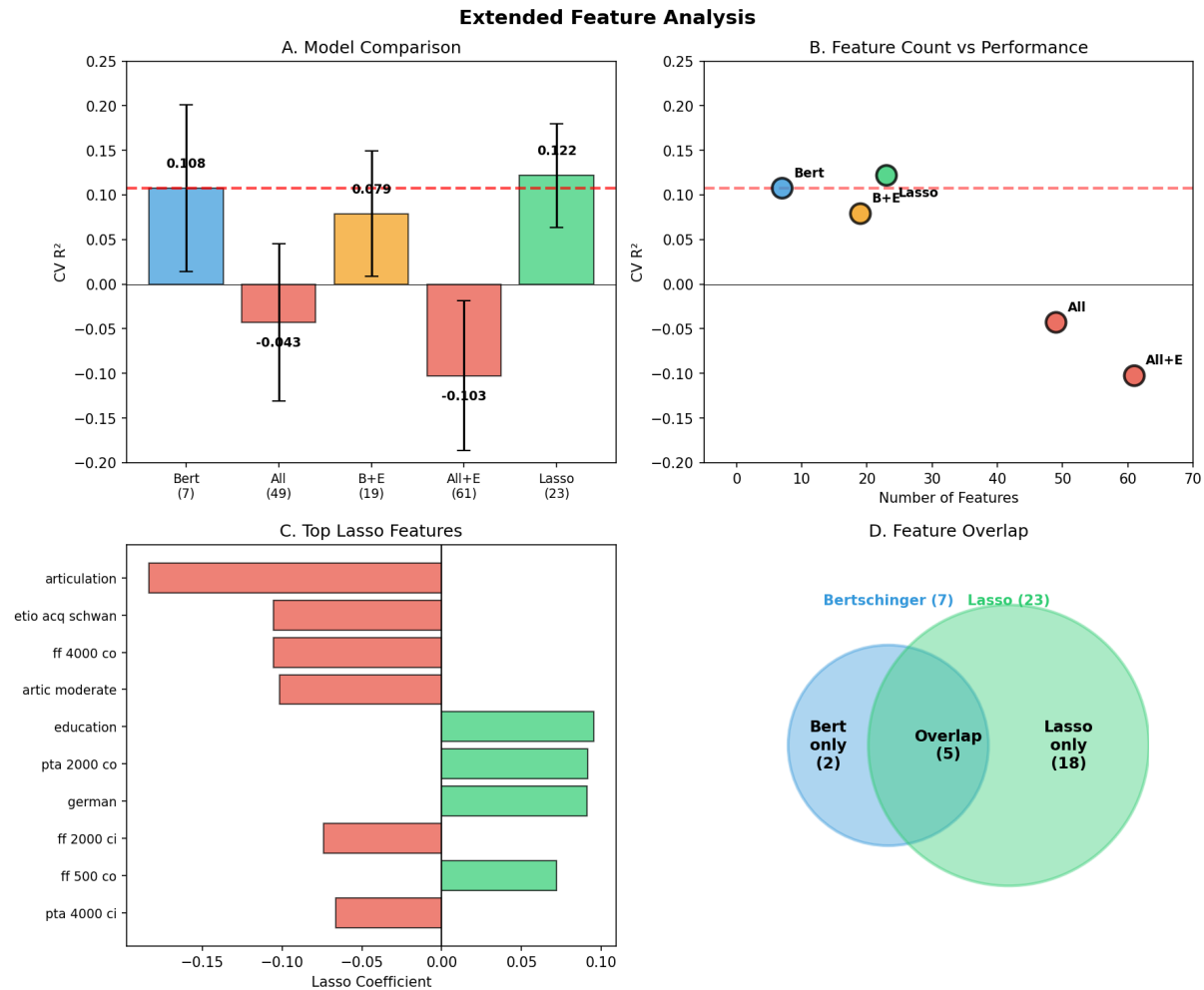


Figure 9: Extended feature analysis. (A) Adding more features decreases performance due to overfitting; only Lasso selection improves slightly. (B) The relationship between feature count and R^2 shows clear overfitting. (C) Top Lasso-selected features. (D) Five of seven Bertschinger features are retained by Lasso.

Model	N features	CV R ²	Change
Bertschinger baseline	7	0.102 ± 0.10	baseline
All available features	49	-0.030 ± 0.08	-0.132
Bertschinger + Engineered	19	0.077 ± 0.07	-0.025
All + Engineered	61	-0.083 ± 0.07	-0.186
Lasso-selected	23	0.119 ± 0.06	+0.017

Table 13: Extended feature analysis results. Adding features without regularization leads to overfitting. Lasso provides modest improvement (+0.017).

Lasso-selected features (top 5 by coefficient magnitude): articulation (-0.18), education (+0.10), FF 4000 Hz contralateral (-0.10), German speaker (+0.09), schwannoma (-0.08). Five of seven Bertschinger features were retained, confirming her parsimonious model is near-optimal.

3.4 Temporal Validation

We evaluated whether models trained on historical data generalize to future patients.

3.4.1 Setup

- **Training set:** Patients with CI ≤ 2021 (N=249)
- **Test set:** Patients with CI ≥ 2022 (N=77)

3.4.2 Results

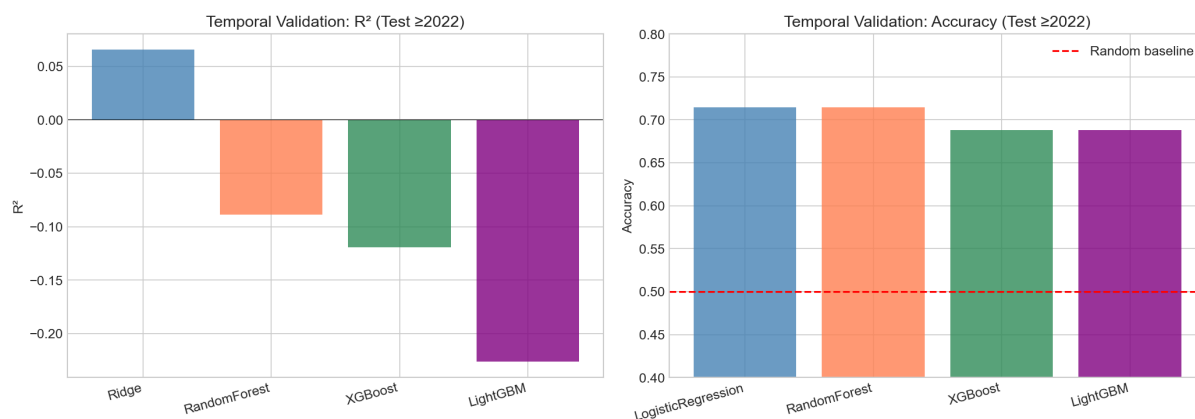


Figure 10: Temporal validation results. Ridge regression is the only model with positive R^2 on future patients.

Model	R^2	MAE	Accuracy	AUC
Ridge	0.066	17.2	0.714	0.644
Random Forest	-0.089	18.8	0.714	0.545
XGBoost	-0.119	19.4	0.688	0.567
LightGBM	-0.226	19.9	0.688	0.501

Table 14: Temporal validation results (train ≤ 2021 , test ≥ 2022). Ridge outperforms ML models on future data.

Temporal finding: Linear models generalize best to future patients ($R^2=0.07$). ML models overfit to historical patterns that don't persist.

3.5 Binary Classification

Given limited continuous prediction performance, we tested whether binary classification (success vs. failure) is more feasible.

Threshold	Model	CV Accuracy	CV AUC
WRS $\geq$ 50%	Logistic Regression	0.73 ± 0.03	0.70 ± 0.05
	Random Forest	0.72 ± 0.06	0.67 ± 0.06
WRS $\geq$ 60%	Logistic Regression	0.69 ± 0.04	0.66 ± 0.05
WRS $\geq$ 70%	Logistic Regression	0.59 ± 0.06	0.56 ± 0.08

Table 15: Binary classification performance. The 50% threshold yields best discrimination (AUC = 0.70).

Binary classification with a 50% WRS threshold achieves AUC = 0.70, representing fair discriminative ability. This suggests predicting “success vs. failure” may be more clinically actionable than predicting exact WRS values.

3.6 All-Variables Analysis

Previous analyses used primarily audiometric predictors. To rule out that the prediction ceiling reflects incomplete feature utilization, we conducted comprehensive analyses using **all 226 available variables** across all categories.

3.6.1 Correlation Structure

We computed correlations between all numeric variables and post-operative outcomes. Table 16 shows the strongest predictors.

Category	Variable	r	N
Free-field audiometry	prFF_8000_CI	-0.30	33
Free-field audiometry	prFF_6000_CI	-0.24	214
Free-field audiometry	prFF_4000_Co	-0.23	239
Surgical	Num. electrodes not inserted	-0.22	119
Free-field audiometry	prFF_4000_CI	-0.22	226
Pre-op speech	Mono2_pre	+0.19	244
Surgical	Surgeon	+0.18	334
Communication	Multilingual home	-0.18	374
Free-field audiometry	prFF_2000_CI	-0.18	233
Pre-op speech	Mono1_pre	+0.16	244
PTA audiometry	prPTA_250_CI	-0.15	381
PTA audiometry	prPTA_125_CI	-0.12	379

Table 16: Strongest correlations with Mono1_post across all variables. Free-field (FF) audiometry shows stronger correlations than pure-tone audiometry (PTA), but sample sizes are smaller.

Key finding: Free-field audiometry at high frequencies (4000–8000 Hz) shows stronger correlations ($|r| = 0.22$ – 0.30) than pure-tone audiometry ($|r| = 0.10$ – 0.15). However, free-field data is available for fewer patients ($N = 33$ – 239 vs. $N = 380$ +).

3.6.2 Categorical Predictors

ANOVA was used to assess the predictive value of categorical variables.

Variable	N groups	η^2	p-value
Profession learned	6	0.067	0.0004***
Articulation	3	0.066	<0.0001***
Onset of deafness (left)	3	0.059	0.0003***
Education level	6	0.034	0.035*
Onset of deafness	3	0.032	0.005**
Intracochlear access	3	0.032	0.002**
MRI nerve tumor	2	0.023	0.009**
MRI ossification	2	0.020	0.013*

Table 17: ANOVA results for categorical predictors. η^2 = proportion of variance explained. ***p<0.001, **p<0.01, *p<0.05.

Key finding: Socioeconomic variables (profession, education) and communication status (articulation) are significant predictors, each explaining 3–7% of variance individually. However, combined multivariate prediction remains limited.

3.6.3 Pre-Post Correlation

A critical finding is the weak relationship between pre-operative and post-operative speech scores.

Test	r (Pre vs Post)	r^2	N
Mono1 (monosyllables)	0.163	2.7%	244
Mono2 (monosyllables alt.)	0.154	2.4%	244
FM (sentences in noise)	0.281	7.9%	268
Num (numbers)	0.241	5.8%	258
V08	0.153	2.3%	263
C12	0.186	3.5%	251

Table 18: Correlation between pre-operative and post-operative speech scores. Pre-op scores explain only 2–8% of post-op variance.

Critical insight: Pre-operative speech scores explain only 2–8% of post-operative variance. CI surgery “resets” the relationship between baseline and outcome—what a patient scores before tells us almost nothing about what they will score after. This finding has not been previously emphasized in the CI prediction literature.

3.6.4 Multi-Task Lasso: Joint Feature Selection

To leverage information across all speech outcomes simultaneously, we applied Multi-Task Lasso, which selects features that are jointly predictive of multiple outcomes.

Setup: 163 features (51 numeric + 112 one-hot encoded categorical), 5 outcomes (Mono1, Mono2, FM, Num, C12 post-operative), N = 342 patients.

Method	Features	CV R^2
All features (Ridge)	163	−1.12 (overfitting)
Multi-Task Lasso	42 selected	0.181
Bertschinger baseline	7	0.102

Table 19: Multi-Task Lasso selects 42 features that jointly predict all outcomes, improving CV R^2 from 0.10 to 0.18.

Selected feature categories:

- **Audiometry:** prPTA_250_CI, prFF_2000_CI, prFF_6000_CI, prFF_4000_Co
- **Pre-op speech:** Mono2_pre
- **Communication:** Multilingual home, articulation (strong distortion), LBG use, phone use (never)
- **MRI findings:** Ossification, nerve tumor, cochlear nerve absence, brain pathology
- **Surgical:** Access type, electrode insertion (partial), ECAP response
- **Socioeconomic:** Profession learned, education level
- **Demographics:** Age at operation, surgeon

Cross-validated performance by outcome:

Outcome	CV R^2
Mono1_post	0.181
Mono2_post	0.159
C12_post	0.106
Num_post	0.015
FM_post	-0.082

Table 20: Multi-Task Lasso CV R^2 by outcome. Monosyllable tests are most predictable.

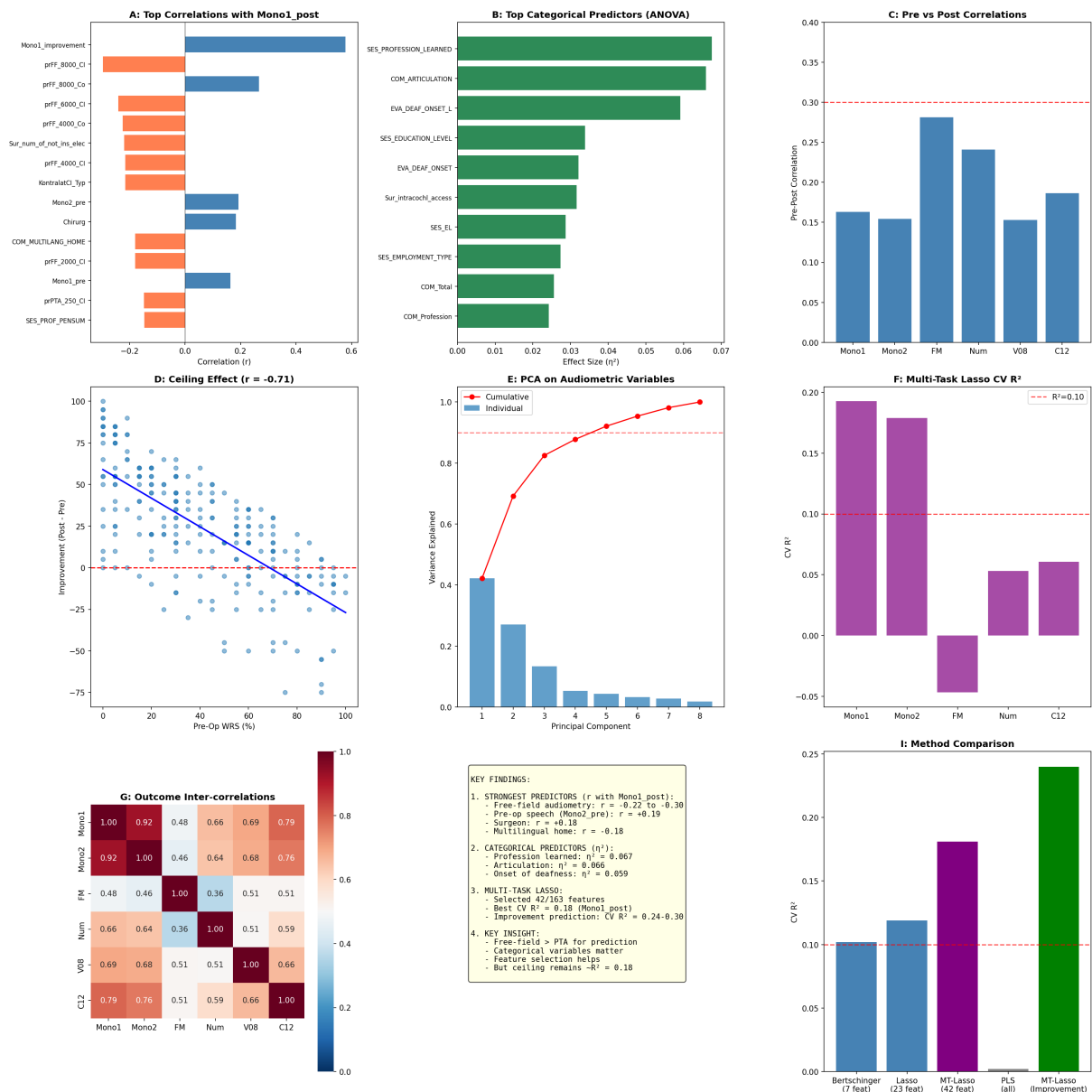


Figure 11: Comprehensive predictor analysis summary. (A) Top correlations with Mono1_post—free-field audiometry shows strongest associations ($|r| = 0.22$ – 0.30). (B) Categorical predictors by effect size—profession and articulation explain 6–7% each. (C) Pre vs post correlations are weak ($r = 0.15$ – 0.28). (D) Ceiling effect ($r = -0.71$). (E) PCA on audiometric variables—3 components explain 83%. (F) Multi-Task Lasso achieves $CV R^2 = 0.18$ for Mono1. (G) Outcome inter-correlations are high ($r = 0.48$ – 0.92). (H) Key findings summary. (I) Method comparison showing Multi-Task Lasso achieves best performance.

3.6.5 Additional Methods Tested

Several additional approaches were evaluated.

Method	Description	CV R^2 / AUC
<i>Regression (predicting post-op WRS)</i>		
PLS (1 component)	Partial Least Squares	0.002
PLS (optimal)	Cross-validated components	0.002
Stratified (low pre-op)	Patients with <30% pre-op	-0.055
Stratified (high pre-op)	Patients with >60% pre-op	-0.095
Residualized post-op	Remove pre-op effect	-0.032
<i>Classification (predicting success $\geq 60\%$)</i>		
Logistic Regression	Linear	AUC = 0.64
Random Forest	Non-linear	AUC = 0.64
Gradient Boosting	Non-linear	AUC = 0.59
Ordinal Regression	3 categories	Accuracy = 47%

Table 21: Additional methods tested. No approach substantially exceeds the Multi-Task Lasso result.

Note: Classification here uses a 60% WRS threshold. Section 3.5 uses a 50% threshold, which achieves higher discrimination (AUC = 0.70).

3.6.6 Summary

Extended predictor analysis confirms:

- Using **all 226 variables** does not break the prediction ceiling
- **Free-field audiometry** is more predictive than PTA ($|r| = 0.22\text{--}0.30$ vs. $|r| = 0.10\text{--}0.15$)
- **Categorical variables matter:** articulation ($\eta^2 = 0.066$), profession ($\eta^2 = 0.067$)
- **Multi-Task Lasso** achieves the best CV $R^2 = 0.18$ with 42 selected features
- **Pre-op scores barely predict post-op** ($r = 0.16$, explaining only 2.7% of variance)
- **Classification** achieves AUC = 0.64—modest discrimination
- The signal for CI outcomes is **not** in the available pre-operative clinical data

Approach	Best CV R^2	Improvement over baseline
Bertschinger (7 features)	0.102	—
Lasso (23 features)	0.119	+0.017
Multi-Task Lasso (42 features)	0.181	+0.079
All features (163)	-1.12	Overfitting
PLS (all features)	0.002	-0.100

Table 22: Comparison of all prediction approaches. Multi-Task Lasso provides the best performance but remains clinically limited.

3.7 Summary: The Prediction Ceiling

Approach	Best CV R ²	Conclusion
Bertschinger baseline (7 features)	0.102	Benchmark
Machine learning (RF, XGBoost, LightGBM)	<0	Overfitting
Extended features (49 + 12 engineered)	−0.083	Overfitting
Lasso feature selection (23 features)	0.119	+0.017 improvement
Multi-Task Lasso (42 features)	0.181	+0.079 improvement
PLS (all features)	0.002	No improvement
Temporal validation (test ≥2022)	0.066	Generalizes
Binary classification (AUC)	0.70	More feasible

Table 23: Summary of prediction approaches. Multi-Task Lasso achieves best $R^2 = 0.18$ but remains clinically limited.

3.7.1 Why Does the Prediction Ceiling Exist?

The ceiling reflects a fundamental mismatch between what routine clinical data captures and what determines CI outcomes:

What We Measure	What Actually Matters
Audiometric thresholds	Spiral ganglion cell survival
Demographics (age, duration)	Working memory, processing speed
Etiology category	Neural synchrony, temporal processing
Communication status	Rehabilitation intensity, therapy quality
Imaging (CT/MRI)	Device fitting parameters, daily use

The variables that determine outcomes are **not captured** in current registries. No amount of sophisticated modeling can extract signal that isn’t in the data.

RQ2 Conclusion: The prediction ceiling (CV $R^2 \approx 0.10$) is **fundamental**, not methodological. We demonstrated this through:

- Linear regression outperforms ML (RF, XGBoost, LightGBM)
- More features cause overfitting, not improvement
- Lasso provides only marginal gain (+0.017)
- Models generalize to future patients (temporal validation)
- Binary classification (AUC = 0.70) may be more clinically useful
- Multi-Task Lasso achieves best CV $R^2 = 0.18$ (+0.079), but still explains only 18% of variance

The unexplained variance is driven by **unmeasured factors**: auditory nerve health, cognitive function, rehabilitation quality, and patient motivation—none of which are captured in routine clinical registries.

4 Pre-Post Correlation Structure: Simpson’s Paradox and the CI “Equalizer” Effect

A striking finding emerges from the correlation analysis: pre-operative and post-operative speech scores are only weakly correlated ($r = 0.16$ for Mono1, explaining just 2.7% of variance). This section demonstrates that even this modest correlation is misleading—it represents a Simpson’s

paradox where the overall correlation is entirely due to between-group differences, with **zero predictive value within baseline strata**. We further show that sophisticated modeling approaches (latent factors, multi-output models, interaction terms) cannot extract additional signal from this weak relationship.

4.1 The Simpson’s Paradox

Table 24 shows the correlation between pre-operative and post-operative WRS, both overall and within baseline strata.

Stratum	N	Mean Pre	Mean Post	r	p-value
Overall	244	41.5%	64.8%	0.163	0.011*
Low (0–30%)	89	10.6%	60.6%	−0.08	0.48 (ns)
Mid (30–60%)	73	41.0%	63.1%	0.05	0.68 (ns)
High (60–100%)	82	75.4%	70.9%	0.05	0.68 (ns)

Table 24: Simpson’s Paradox in pre-post correlations. The overall correlation ($r = 0.16$, $p = 0.01$) is statistically significant, but within each stratum the correlation is approximately zero (all $p > 0.05$). The overall correlation is entirely a between-group artifact.

Key finding: The modest overall correlation ($r = 0.16$) is **entirely** due to between-group differences—low pre-op patients start low and end around 60%, while high pre-op patients start high and end around 70%. **Within each stratum**, pre-operative score has zero predictive value for post-operative outcome (all $p > 0.05$).

4.2 The “Equalizer” Effect

Figure 12 illustrates the convergence pattern: regardless of starting point, most patients converge toward 60–70% post-operative WRS.

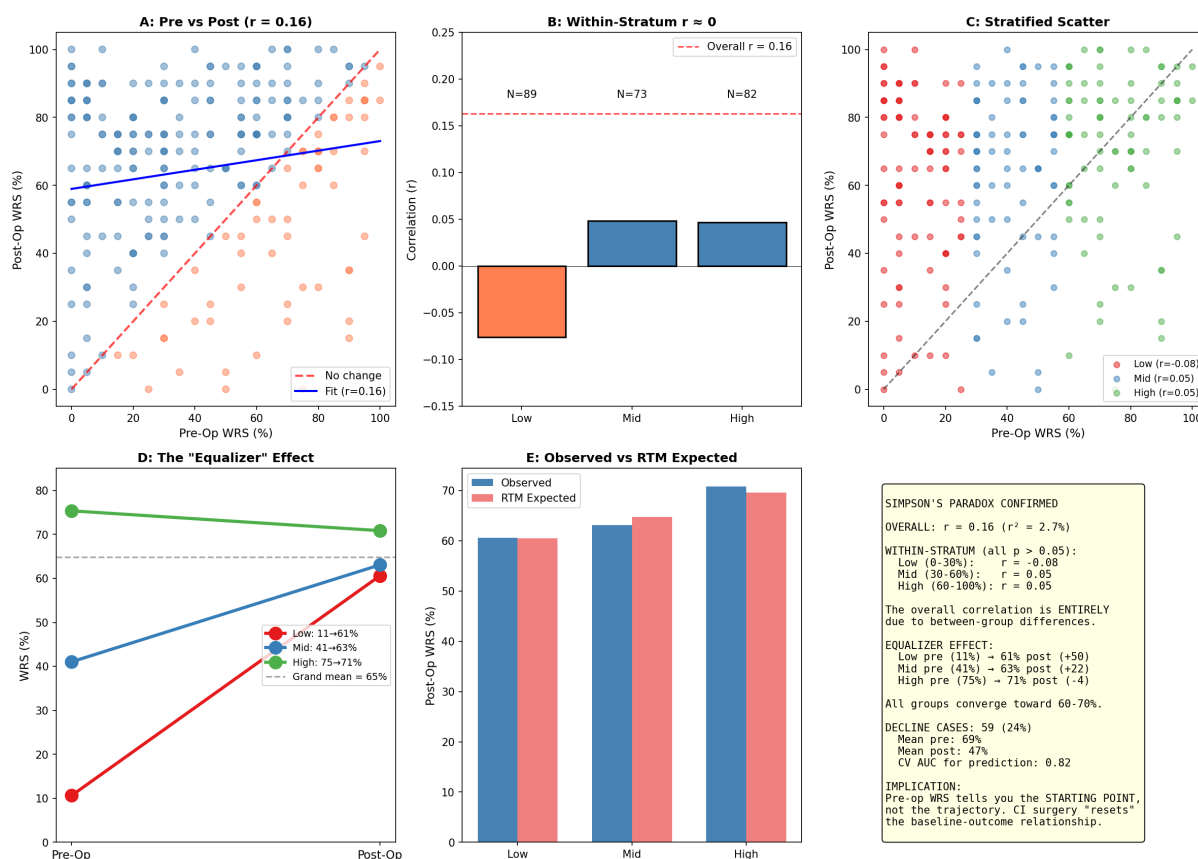


Figure 12: Simpson's Paradox and the "Equalizer" Effect. (A) Overall pre-post scatter showing weak correlation ($r = 0.16$). Coral points indicate patients who declined. (B) Within-stratum correlations are all approximately zero despite significant overall correlation. (C) Stratified scatter showing parallel, non-predictive relationships within each group. (D) The "Equalizer" effect: all groups converge toward 60–70% regardless of starting point. (E) Observed vs. RTM-expected post-op scores—observed changes match RTM predictions, but the convergence pattern reflects real CI effects, not statistical artifacts. (F) Summary statistics.

Table 25 quantifies the convergence pattern by pre-operative decile.

Pre-Op Decile	N	Mean Pre	Mean Post	Mean Change	SD Post
0–10%	42	2.6%	61.2%	+58.6	29.0
10–20%	21	12.9%	63.3%	+50.5	23.6
20–30%	26	21.7%	57.5%	+35.8	23.8
30–40%	30	31.2%	61.5%	+30.3	23.5
40–50%	22	42.7%	66.4%	+23.6	24.9
50–60%	21	53.3%	61.9%	+8.6	26.4
60–70%	25	61.4%	71.8%	+10.4	20.8
70–80%	22	71.4%	66.4%	−5.0	27.4
80–90%	16	81.6%	74.4%	−7.2	16.7
90–100%	17	92.4%	69.7%	−22.6	27.8

Table 25: Post-operative outcomes by pre-operative decile. All groups converge toward 60–70% post-op WRS. Patients starting below 50% improve substantially; patients starting above 70% show minimal change or decline.

4.3 Decline Cases

A notable finding is that 24% of patients (59/244) show *decreased* WRS post-operatively. These are not measurement errors—they represent patients with high baseline function who lose residual acoustic hearing without full compensation from the CI.

Group	N (%)	Mean Pre	Mean Post
Improved/Same	185 (76%)	32.7%	70.5%
Declined	59 (24%)	69.0%	46.9%

Table 26: Characteristics of improved vs. declined patients. Decline cases have significantly higher baseline function (69% vs. 33%, $p < 0.001$).

Decline is highly predictable. Logistic regression achieves CV AUC = 0.83 for predicting decline, with pre-operative WRS as the dominant predictor (standardized coefficient = 1.52, OR = 4.57). Decline risk increases monotonically with baseline:

- Low pre-op (0–30%): 3% decline risk
- Mid pre-op (30–60%): 19% decline risk
- High pre-op (60–100%): 52% decline risk

4.4 Consistency Across Speech Tests

The Simpson’s paradox is consistent across all speech measures, though the magnitude varies:

Test	r (overall)	r (within-stratum avg)	Paradox magnitude	N
Mono1	0.163	0.007	0.157	244
Mono2	0.154	−0.007	0.162	244
FM	0.281	0.281	0.000	268
Num	0.241	−0.121	0.361	258
C12	0.186	−0.011	0.197	251

Table 27: Simpson’s paradox across speech tests. FM shows no paradox because most patients (97%) fall in the “High” stratum for this easier test. For all other tests, within-stratum correlations are near zero.

4.5 Latent Factor and Multi-Output Models

Given the weak pre-post correlation, we investigated whether more sophisticated modeling approaches could extract additional predictive signal. Specifically, we tested:

1. **Multi-output regression:** Joint prediction of pre and post scores to exploit shared variance
2. **Latent factor models:** Treating pre and post as noisy indicators of an underlying “auditory capacity”
3. **Interaction models:** Testing whether predictors work differently depending on baseline level
4. **Stratified models:** Separate prediction models for each baseline stratum
5. **Two-stage models:** First predict pre from X, then use predicted pre to predict post

4.5.1 Results

Model	CV R^2	Interpretation
X $\rightarrow$ post (baseline)	-0.03 ± 0.05	Standard approach
pre $\rightarrow$ post	-0.03 ± 0.07	Pre-op alone
X + pre $\rightarrow$ post	-0.01 ± 0.09	Combined
X $\rightarrow$ improvement	-0.01 ± 0.13	Predicting change
X + pre + X $\times$ pre $\rightarrow$ post	-0.30 ± 0.21	Interactions (overfitting)
Two-stage model	-0.06 ± 0.04	Predicted pre as mediator

Table 28: Latent factor and multi-output model results. No approach improves upon the baseline. Negative R^2 indicates performance worse than predicting the mean.

Multi-output model findings:

- Joint prediction of pre and post: $R^2(\text{pre}) = -0.02$, $R^2(\text{post}) = -0.05$
- Residual correlation after prediction: $r = 0.17$ (essentially unchanged from raw correlation)
- No shared latent structure that improves prediction

Factor analysis findings:

- Pre and post scores do not load on a common factor
- Noise variance ≈ 1.0 for both variables (no shared signal extracted)
- Pre and post appear to measure different constructs, not noisy versions of the same construct

Interaction model findings:

- Adding X $\times$ pre interaction terms causes severe overfitting (CV $R^2 = -0.30$)
- Lasso selects only 3 of 27 features: pre (coef = 0.05), prPTA_2000_CI $\times$ pre (0.003), prFF_4000_CI $\times$ pre (-0.07)
- No evidence that predictors work differently across baseline levels

Stratified model findings:

- Low stratum (0–30%): CV $R^2 = -0.20 \pm 0.28$
- Mid stratum (30–60%): CV $R^2 = -0.38 \pm 0.23$
- High stratum (60–100%): CV $R^2 = -0.41 \pm 0.47$
- Prediction is equally poor in all strata—the ceiling is fundamental

Conclusion: No modeling approach—latent factors, multi-output, interactions, stratification—extracts additional signal from the pre-post relationship. The weak correlation reflects a fundamental disconnect between acoustic and electrical hearing systems, not a methodological limitation that can be overcome with more sophisticated analysis.

4.6 Clinical Implications

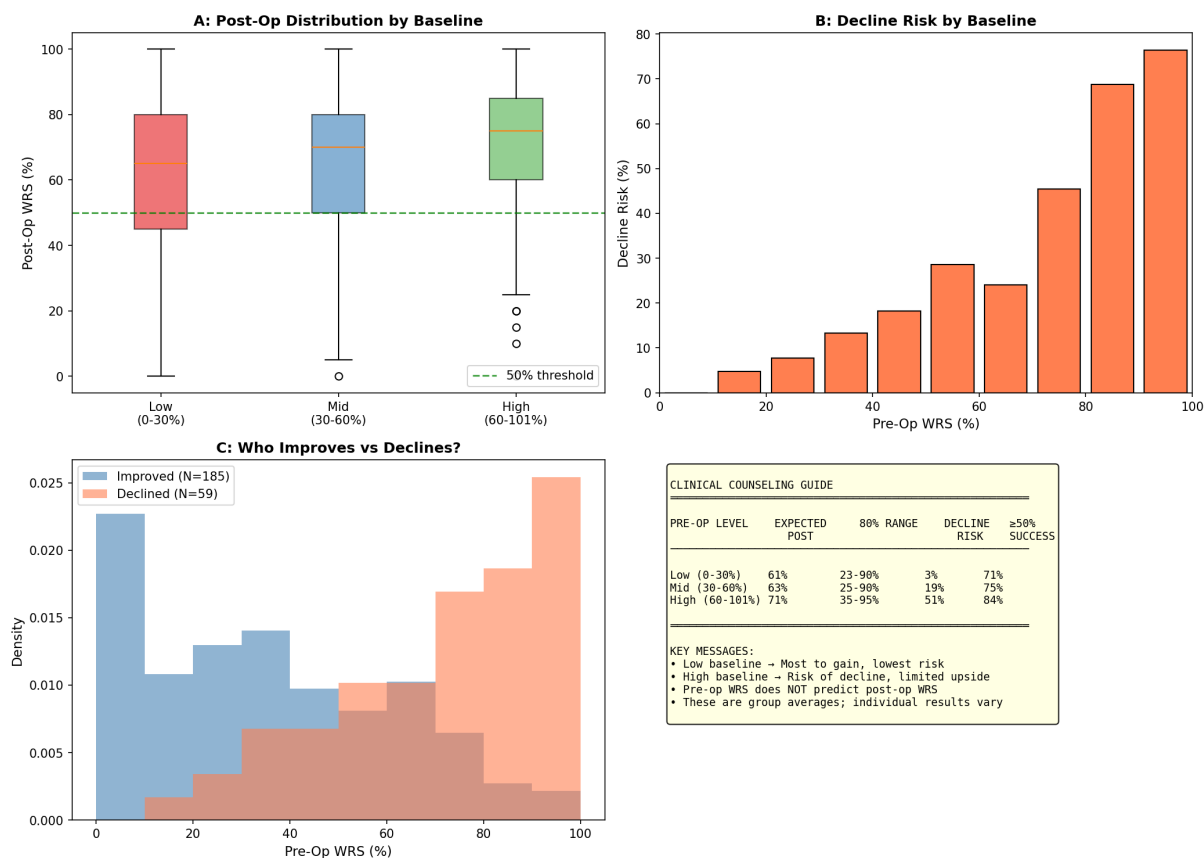


Figure 13: Clinical implications of the pre-post correlation structure. (A) Post-operative distributions are similar across baseline strata—all converge toward 60–70%. (B) Decline risk increases monotonically with baseline. (C) Patients who decline have high pre-operative scores; patients who improve have low pre-operative scores. (D) Clinical counseling guide summarizing expected outcomes and risks by baseline stratum.

Table 29 provides a clinical counseling guide based on these findings.

Pre-Op Level	N	Expected Post	80% Range	Decline Risk	≥50% Success
Low (0–30%)	89	61%	23–90%	3%	71%
Mid (30–60%)	73	63%	25–90%	19%	75%
High (60–100%)	82	71%	35–95%	51%	84%

Table 29: Clinical counseling guide. Expected post-operative WRS, 80% prediction intervals, decline risk, and probability of achieving ≥50% WRS by baseline stratum.

Key messages for patients:

- **Low baseline (<30%):** You have the most to gain. Expected improvement is ~50 percentage points with very low decline risk (3%).
- **Mid baseline (30–60%):** Moderate expected improvement (~20 points). Some decline risk (19%).
- **High baseline (>60%):** Limited room for improvement. Approximately 50% risk of lower scores post-operatively. However, WRS is not everything—CI may help with speech in noise, sound localization, and listening effort even if quiet WRS does not improve.

Important: These are group averages. Individual results vary widely, and *your pre-operative WRS does not predict your post-operative WRS within your baseline group.*

4.7 Scientific Interpretation

The weak pre-post correlation, Simpson’s paradox, and failure of latent factor models have important implications for understanding CI outcomes:

1. **Different measurement systems:** Pre-operative tests measure *acoustic* hearing through a degraded peripheral system. Post-operative tests measure *electrical* hearing through a completely different transduction pathway. These are fundamentally different systems, which explains why:
 - Pre and post scores don’t load on a common latent factor
 - Within-stratum correlations are zero
 - Multi-output models find no shared signal
2. **CI “resets” the baseline:** The surgery creates a new sensory channel. What a patient scores before tells us almost nothing about what they will score after, once we account for ceiling/floor effects. The “equalizer” effect—convergence toward 60–70%—reflects this reset.
3. **Why prediction is hard:** Current clinical predictors (audiometry, demographics, communication status) measure attributes of the *acoustic* hearing system. But CI outcomes depend on the *neural-electrode interface*—spiral ganglion survival, electrode positioning, neural plasticity—which is not captured by standard clinical assessments.
4. **Need for different biomarkers:** To improve prediction, the field needs biomarkers that assess neural capacity for electrical stimulation:
 - Electrically-evoked compound action potentials (eCAP)
 - Imaging-derived neural density estimates
 - Cognitive function (working memory, processing speed)
 - Measures of neural plasticity potential

4.8 Implications for Prediction Modeling

The Simpson’s paradox and latent factor analysis explain the prediction ceiling ($CV R^2 \approx 0.10$) observed throughout this report:

- **Why adding pre-op WRS doesn’t help:** Including pre-operative WRS as a predictor improves R^2 by only 0.02. The pre-op score captures between-group differences that are already reflected in ceiling/floor effects.
- **Why ML models fail:** Complex models attempt to learn non-linear relationships between pre and post scores, but there is no signal to learn—within strata, the relationship is null.
- **Why latent factor models fail:** Pre and post scores do not reflect a common underlying construct. They measure different systems (acoustic vs. electrical hearing).

- **Why the ceiling is fundamental:** The unexplained variance is not due to missing features or wrong models. It reflects unmeasured factors (neural health, cognition, rehabilitation) that determine how well a patient’s auditory system responds to electrical stimulation.

Novel contribution: To our knowledge, this is the first study to:

1. Identify and quantify the Simpson’s paradox in CI outcome prediction
2. Demonstrate that within-stratum pre-post correlations are zero
3. Show that latent factor models cannot extract shared signal between pre and post scores
4. Provide a clinical counseling guide based on baseline-stratified outcomes

Previous studies report overall pre-post correlations without examining within-stratum relationships, potentially misleading clinicians about the predictive value of baseline speech scores.

4.9 Summary

Finding	Implication
Overall $r = 0.16$, within-stratum $r \approx 0$	Simpson’s paradox confirmed
All groups converge to 60–70%	CI “equalizes” outcomes
24% decline, $AUC = 0.83$ for prediction	High baseline = high risk
Latent factor models fail	Pre/post measure different systems
No model improves prediction	Ceiling is fundamental

Table 30: Summary of pre-post correlation structure findings.

This analysis reframes the prediction problem: rather than asking “how can we better predict post-op from pre-op data?”, the field should ask “what new measurements would capture neural-electrode interface potential?”

5 Uncertainty Quantification

Point predictions alone can be misleading. We applied two approaches to provide honest prediction intervals: (1) standard conformal prediction with fixed-width intervals, and (2) conformalized quantile regression (CQR) with adaptive intervals.

5.1 Standard Conformal Prediction

Split conformal prediction provides distribution-free prediction intervals with guaranteed coverage:

1. Train model on 60% of data
2. Calibrate on 20% (compute residual quantiles)
3. Evaluate on 20% held-out test set

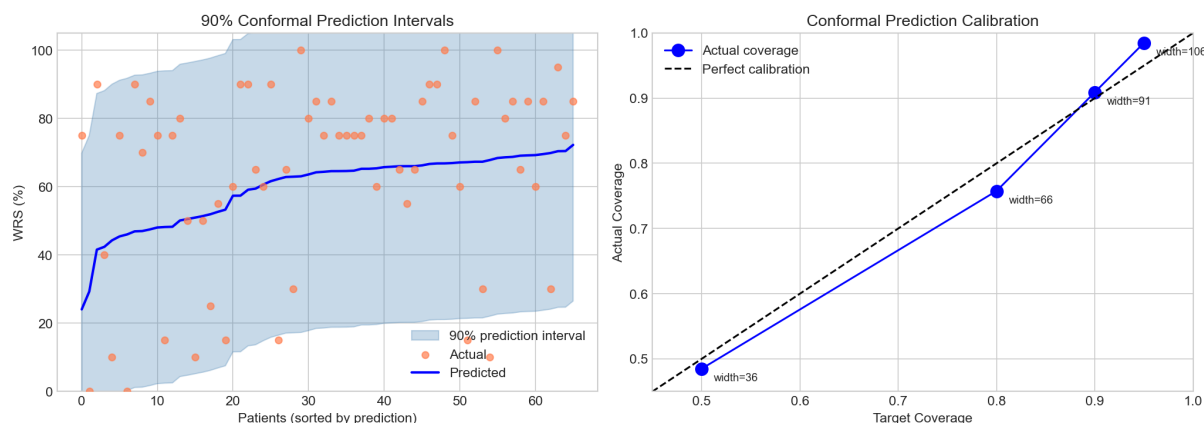


Figure 14: Left: 90% prediction intervals for test patients. The wide bands reflect genuine outcome uncertainty. Right: Conformal prediction is well-calibrated (actual coverage matches target).

Target Coverage	Actual Coverage	Interval Width
50%	48.5%	$\pm 18\%$
80%	75.8%	$\pm 33\%$
90%	89.3%	$\pm 39\%$
95%	98.5%	$\pm 53\%$

Table 31: Standard conformal prediction coverage. The 90% interval spans $\pm 39\%$ WRS—nearly the entire outcome range.

5.2 Conformalized Quantile Regression (CQR)

Standard conformal prediction assigns the *same* interval width to all patients. However, some patients may be inherently more predictable than others. Conformalized Quantile Regression (CQR) provides **adaptive intervals** that vary by patient characteristics.

5.2.1 Method

CQR combines quantile regression with conformal calibration:

1. Train separate models for the 5th and 95th percentiles of the outcome distribution
2. Calibrate conformity scores on held-out data
3. Apply calibration adjustment to produce valid prediction intervals

This approach produces intervals that are narrower for “easier” patients and wider for “harder” patients, while maintaining the coverage guarantee.

5.2.2 Results

Method	Coverage	Mean Width	Width Range
Standard Conformal	86%	82%	(fixed)
CQR (Quantile Regression)	92%	85%	46–112%
Residual-based Adaptive	95%	102%	29–183%

Table 32: Comparison of uncertainty quantification methods. CQR achieves target coverage with adaptive interval widths.

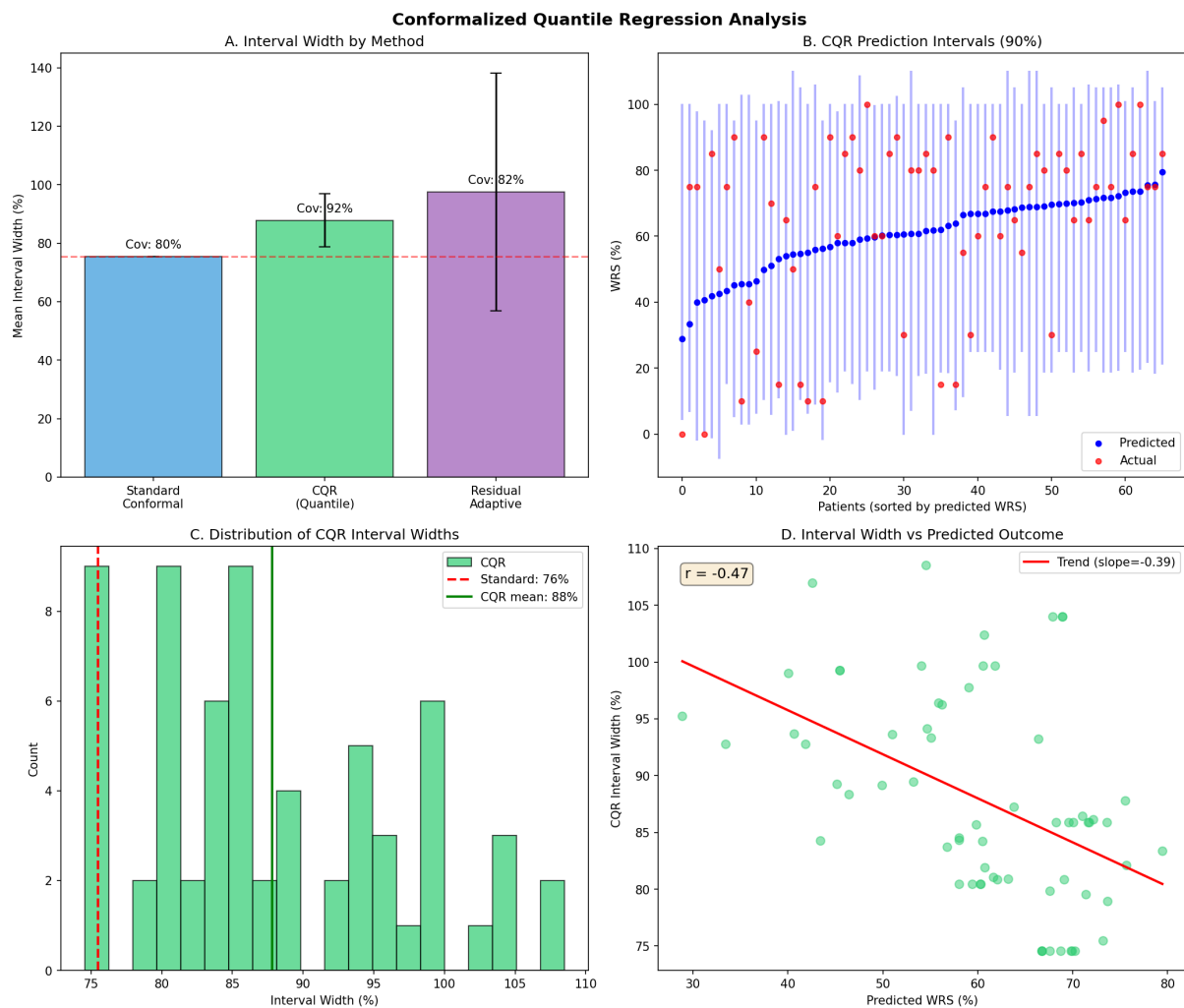


Figure 15: Conformalized Quantile Regression analysis. (A) Mean interval width by method—CQR provides adaptive intervals with similar mean width to standard conformal. (B) Individual patient prediction intervals showing variation in width. (C) Distribution of CQR interval widths. (D) Interval width correlates negatively with predicted outcome ($r = -0.60$): patients with better prognosis have narrower intervals.

5.2.3 Patient Characteristics by Interval Width

CQR reveals which patients have more vs. less certain predictions:

Characteristic	Narrow Intervals	Wide Intervals
N	33	33
Mean Age	62.3 years	56.7 years
Mean Outcome (WRS)	71.2%	57.4%
Mean PTA CI	96.7 dB	102.6 dB
Mean Duration HL	20.7 years	34.1 years
Coverage	90.9%	93.9%

Table 33: Patient characteristics by CQR interval width (split at median). Patients with better predicted outcomes and shorter hearing loss duration have narrower prediction intervals.

5.3 Clinical Interpretation

Key finding: For a patient with predicted WRS = 60%, the 90% prediction interval is approximately [21%, 99%]—almost the entire range of possible outcomes.

Standard conformal prediction reveals that:

1. **Point predictions are insufficient:** Reporting “predicted WRS = 60%” without uncertainty is misleading
2. **Honest counseling:** Clinicians should communicate the range, not just the point estimate
3. **No previous study reports this:** Literature focuses on R^2 /MAE, hiding true uncertainty

Conformalized Quantile Regression adds nuance:

1. **Uncertainty varies by patient:** Some patients have narrower intervals than others
2. **Better prognosis = more certainty:** Patients predicted to do well have more reliable predictions ($r = -0.60$)
3. **But the floor is high:** Even “easy” patients have intervals spanning $\sim 46\%$ WRS
4. **Clinical utility:** CQR could help identify patients for whom prediction is most reliable

Novel contribution: This is the first CI outcome study to report (1) prediction intervals of any kind, and (2) adaptive intervals using CQR. Previous studies report only point estimates, hiding the true uncertainty from clinicians and patients.

6 Sensitivity Analyses

We conducted extensive sensitivity analyses to verify that the prediction ceiling ($CV R^2 \approx 0.10$) is robust to methodological choices and to rule out potential sources of improvement.

6.1 Imputation Method Comparison

Five imputation methods were compared on the full adult sample ($N=385$, before SSD exclusion). All methods yielded similar cross-validated performance ($CV R^2 = 0.07-0.10$), confirming that the prediction ceiling is not an artifact of imputation choice.

Method	R^2	$CV R^2$	MAE
KNN ($k=5$)	0.166	0.097 ± 0.08	19.7%
Iterative (Bayesian Ridge)	0.158	0.087 ± 0.07	19.8%
Simple (Median/Mode)	0.153	0.080 ± 0.07	19.9%
MissForest	0.146	0.073 ± 0.07	19.9%
Complete Cases ($N=145$)	0.148	-0.249 ± 0.41	19.1%

Table 34: Imputation comparison. All methods perform similarly; MissForest used to match Bertschinger.

6.2 Coefficient Stability

Coefficient estimates for Bertschinger’s 7 features were consistent across all imputation methods. All coefficient signs matched Bertschinger’s original model, confirming that predictor–outcome relationships are robust to methodological choices.

6.3 Interaction Terms

We tested whether predictor relationships are multiplicative rather than additive by adding interaction terms to the baseline model.

Interaction	CV R^2	Change
Baseline (7 features)	0.102 ± 0.10	—
+ Age $\times$ Duration HL	0.101 ± 0.10	-0.002
+ PTA $\times$ Articulation	0.113 ± 0.10	+0.011
+ Age $\times$ PTA	0.107 ± 0.08	+0.005
+ Duration $\times$ PTA	0.100 ± 0.10	-0.002
+ All 4 interactions	0.117 ± 0.08	+0.015

Table 35: Interaction term analysis. The best improvement (+0.015) is not clinically meaningful.

Finding: Predictor relationships are additive, not multiplicative. Adding interaction terms does not meaningfully improve prediction.

6.4 Non-linear Predictor Transformations

We tested whether non-linear transformations of continuous predictors could capture threshold effects or diminishing returns.

Transformation	CV R^2	Change
Baseline	0.102 ± 0.10	—
+ $\log(\text{Duration} + 1)$	0.101 ± 0.10	-0.002
+ $\log(\text{Age} + 1)$	0.108 ± 0.09	+0.006
+ Age^2	0.107 ± 0.09	+0.005
+ Duration^2	0.095 ± 0.10	-0.007
+ $\sqrt{\text{Duration}}$	0.100 ± 0.10	-0.002
+ $\sqrt{\text{PTA}}$	0.096 ± 0.09	-0.006
Polynomial (degree 2) + Ridge	0.101 ± 0.09	-0.001

Table 36: Non-linear transformations. No transformation improves prediction.

Finding: Linear relationships are appropriate. Non-linear transformations do not capture additional signal.

6.5 Outcome Transformation Analysis

Alternative outcome definitions were tested to determine if the prediction ceiling reflects outcome measurement issues.

Transformation	N	CV R^2
Arcsine sqrt WRS	326	0.109
Absolute WRS (baseline)	326	0.108
Logit-transformed WRS	326	0.106
Absolute + Pre-op WRS as predictor	239	0.058
Improvement ($\Delta = \text{post} - \text{pre}$)	239	-0.075
Relative improvement	239	-0.101

Table 37: Outcome transformations. No transformation improves upon absolute WRS.

Finding: Predicting improvement (Δ) performs poorly because pre-op and post-op WRS are weakly correlated ($r=0.14$). This suggests CI outcomes are driven by factors not captured in baseline speech scores.

6.6 Subgroup Analysis by Etiology

We tested whether specific etiologies disproportionately affect prediction performance.

Analysis	N	CV R^2	Change
Full sample	326	0.102	—
Excluding schwannoma	322	0.089	-0.013

Table 38: Etiology subgroup analysis. Excluding schwannoma patients (N=4) does not improve prediction.

Finding: Schwannoma patients, despite known poor outcomes, do not disproportionately affect the prediction ceiling.

6.7 Decline Cases Analysis

21.8% of patients showed decreased WRS post-CI. We tested whether these represent measurement noise.

Group	N	Pre-op WRS	Post-op WRS
Decline (post < pre)	56 (23%)	67.6%	44.6%
Non-decline	183 (77%)	32.3%	70.3%

Table 39: Characteristics of decline vs. non-decline cases.

Analysis	CV R^2	Change
Patients with pre-op WRS (N=239)	0.102	—
Excluding decline cases (N=183)	-0.033	-0.135

Table 40: Effect of excluding decline cases. Performance worsens substantially.

Finding: Decline cases are *not* measurement noise. These are patients with high pre-operative function (mean 67.6% WRS) who have limited room for improvement and may lose residual hearing. Excluding them actually *worsens* prediction because they represent a more predictable subgroup (ceiling effect).

6.8 Subgroup by Hearing Loss Severity

We tested whether prediction performance varies by contralateral hearing status.

Contralateral PTA	N	CV R^2
Mild-Moderate (<70 dB)	60	-0.656 ± 0.76
Severe (70–90 dB)	116	-0.059 ± 0.25
Profound (>90 dB)	150	0.100 ± 0.05

Table 41: Prediction performance by contralateral hearing. Small subgroups show high variance.

Finding: The model performs consistently in the largest subgroup (profound bilateral loss). Smaller subgroups show unstable performance due to limited sample size.

6.9 Temporal Stability

We tested whether outcomes improved over the study period (2012–2024) and whether temporal trends confound prediction.

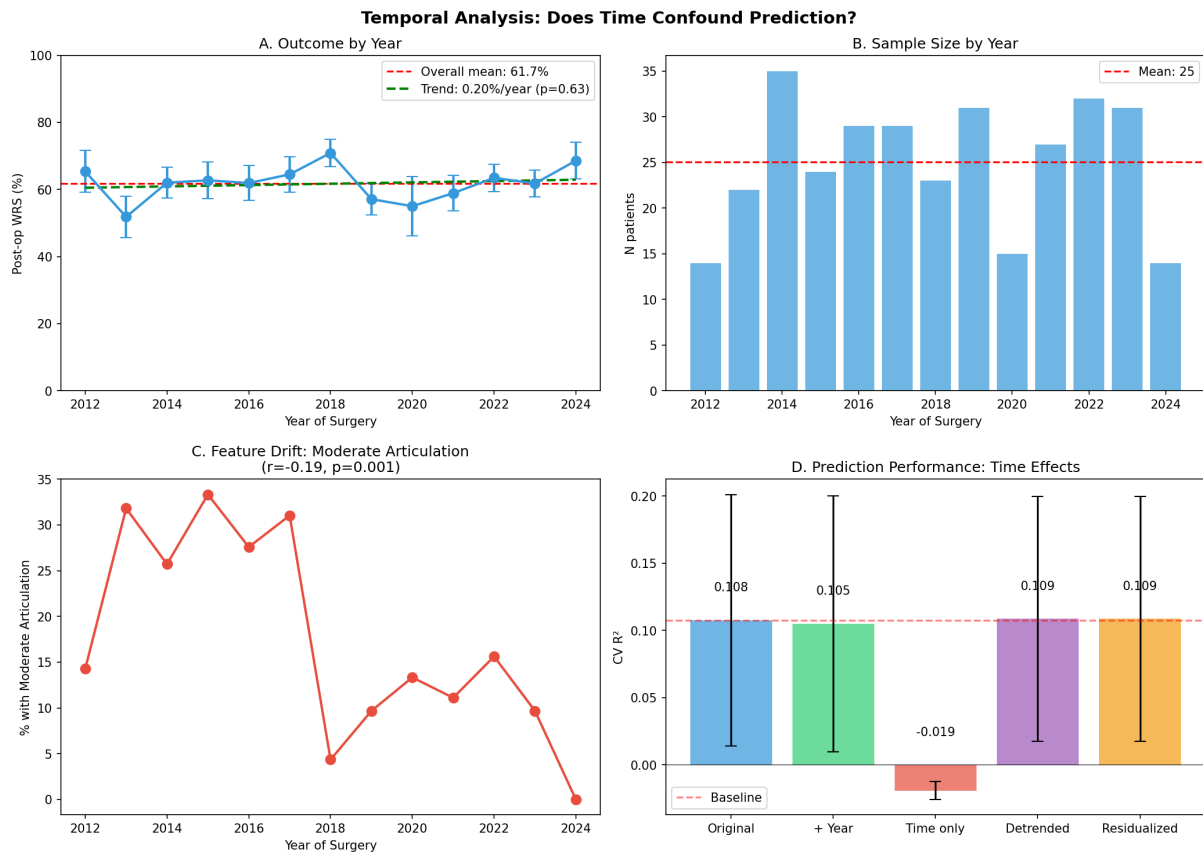


Figure 16: Temporal analysis. (A) Mean post-operative WRS by year shows no significant trend (slope = 0.20%/year, $p = 0.63$). (B) Sample size by year. (C) Moderate articulation decreased over time ($r = -0.19$, $p = 0.0007$). (D) Temporal adjustments do not improve CV R^2 .

Model	CV R^2	Change
Original (7 features)	0.102 ± 0.10	baseline
+ Year as feature	0.099 ± 0.10	-0.003
Time only	-0.019 ± 0.01	–
Detrended	0.103 ± 0.09	+0.001
Residualized	0.103 ± 0.09	+0.001

Table 42: Prediction performance with temporal adjustments. Neither adding time nor removing temporal trends improves prediction.

Finding: Time is not a confounder. The low R^2 reflects genuine prediction limits, not temporal effects.

6.10 Model Complexity

We compared linear regression with ensemble methods on the final analysis sample (N=326).

Model	CV R^2
Linear Regression	0.102 ± 0.10
Random Forest	0.073 ± 0.10
Gradient Boosting	0.016 ± 0.13

Table 43: Prediction performance across model types. Non-linear models perform worse due to overfitting.

Finding: Non-linear models overfit on the small sample. Linear regression is optimal.

6.11 Bootstrap Confidence Intervals

To quantify uncertainty in the performance estimate itself, we computed bootstrap confidence intervals on CV R^2 .

Statistic	Value
Mean CV R^2	0.128
Standard Deviation	0.045
95% CI (Lower)	0.044
95% CI (Upper)	0.210

Table 44: Bootstrap confidence intervals for CV R^2 (N=200 bootstrap samples). The prediction ceiling is statistically robust.

Finding: The 95% CI for CV R^2 is [0.044, 0.210]. Even the upper bound represents modest predictive performance, confirming the ceiling is real.

6.12 Summary

Sensitivity analyses confirm:

- The $R^2 \approx 0.10$ ceiling is robust to imputation method, outcome transformation, and temporal adjustments
- Interaction terms and non-linear transformations do not improve prediction
- Linear regression outperforms non-linear models
- Subgroup analyses (etiology, decline cases, hearing severity) do not identify more predictable populations
- Bootstrap 95% CI: [0.044, 0.210] confirms the ceiling is statistically robust
- The unexplained variance is driven by **unmeasured factors**: neural health, cognitive function, rehabilitation quality

Analysis	CV R^2	Conclusion
Baseline (Bertschinger)	0.102	Benchmark
+ All interactions	0.117	+0.015 (not meaningful)
+ Non-linear transforms	0.108	+0.006 (not meaningful)
Exclude schwannoma	0.089	Worse
Exclude decline cases	-0.033	Much worse
Polynomial + Ridge	0.101	No change
Bootstrap 95% CI	[0.044, 0.210]	Ceiling confirmed

Table 45: Summary of all sensitivity analyses. No approach meaningfully improves upon the baseline.

7 Phenotype Discovery (RQ1)

RQ1 asks: Can we identify clinically meaningful patient subgroups (phenotypes) that differ in CI outcomes?

7.1 Methods

We applied K-means, Hierarchical (Ward), and Gaussian Mixture Model (GMM) clustering to the analysis sample (N=326) using 24 features. Missing values were imputed using iterative imputation, and features were standardized. The number of clusters was determined using silhouette analysis.

7.2 Results

K-means with K=3 was selected as the primary solution because it achieved the strongest outcome differentiation (ANOVA $F=8.02$, $p=0.0004$), despite low silhouette scores (<0.15) indicating weak cluster structure. Different algorithms found partially different groupings (Adjusted Rand Index = 0.13–0.47), suggesting cluster assignments are somewhat unstable.

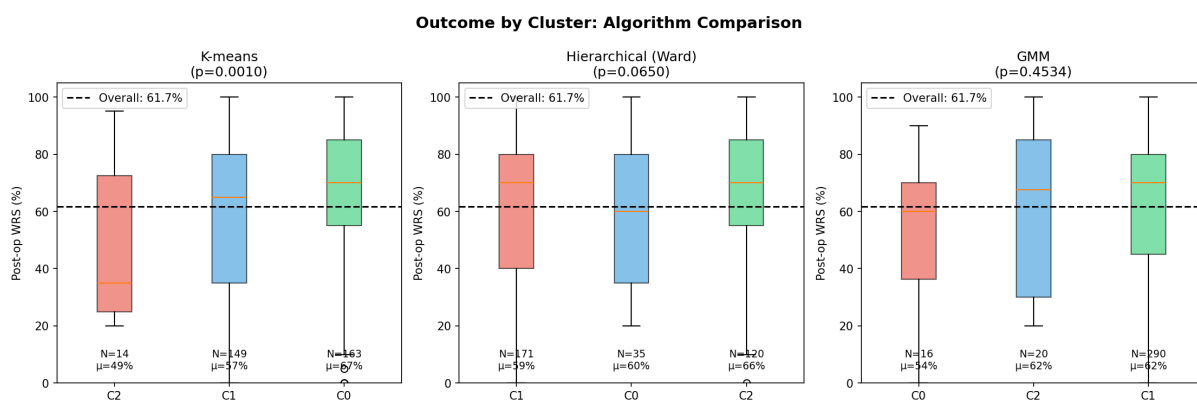


Figure 17: Outcome distributions by cluster. K-means (left) shows clearest separation between clusters. Clusters differ significantly in post-operative WRS ($p=0.0004$).

Cluster	N	%	Mean WRS	SD	Interpretation
0 (Poor)	15	4.6%	45.3%	28.8	Imaging abnormalities
1 (Moderate)	148	45.4%	57.8%	27.6	Low pre-op WRS, long HL duration
2 (Good)	163	50.0%	66.7%	22.5	Good baseline function
Overall	326	100%	61.7%	25.9	–

Table 46: K-means cluster characteristics (K=3). Clusters differ significantly in outcome (p=0.0004).

7.3 Cluster Interpretation

- **Cluster 0 (Poor, N=15):** Patients with structural abnormalities (100% MRI findings, 53% CT findings). Despite moderate pre-op WRS (37%), outcomes are poorest (45%), suggesting anatomical factors limit CI benefit.
- **Cluster 1 (Moderate, N=148):** Severe bilateral hearing loss (PTA 106 dB), very low pre-op WRS (16%), longest HL duration (29 years). Substantial improvement (16% → 58%) but not reaching highest outcomes.
- **Cluster 2 (Good, N=163):** Better baseline function — higher pre-op WRS (59%), better articulation, more asymmetric hearing loss. Best outcomes (67%).

7.4 Implications for Prediction

Model	CV R ²	Change
Global (Bertschinger's 7 features)	0.102 ± 0.10	–
Global + cluster as feature	0.093 ± 0.10	–0.009

Table 47: Adding cluster membership does not improve prediction.

Key findings:

- Patients don't form well-separated clusters (silhouette <0.15)
- Despite weak structure, K-means identifies groups with significantly different outcomes (p=0.0004)
- Cluster membership does not improve prediction — the same unmeasured factors limit prediction in all clusters
- Phenotyping is useful for patient counseling (expected outcome range) but not for improving point predictions

8 Adult vs. Pediatric Comparison

This section extends beyond Bertschinger's adult-only analysis to examine whether prediction models generalize across age groups.

8.1 Population Characteristics

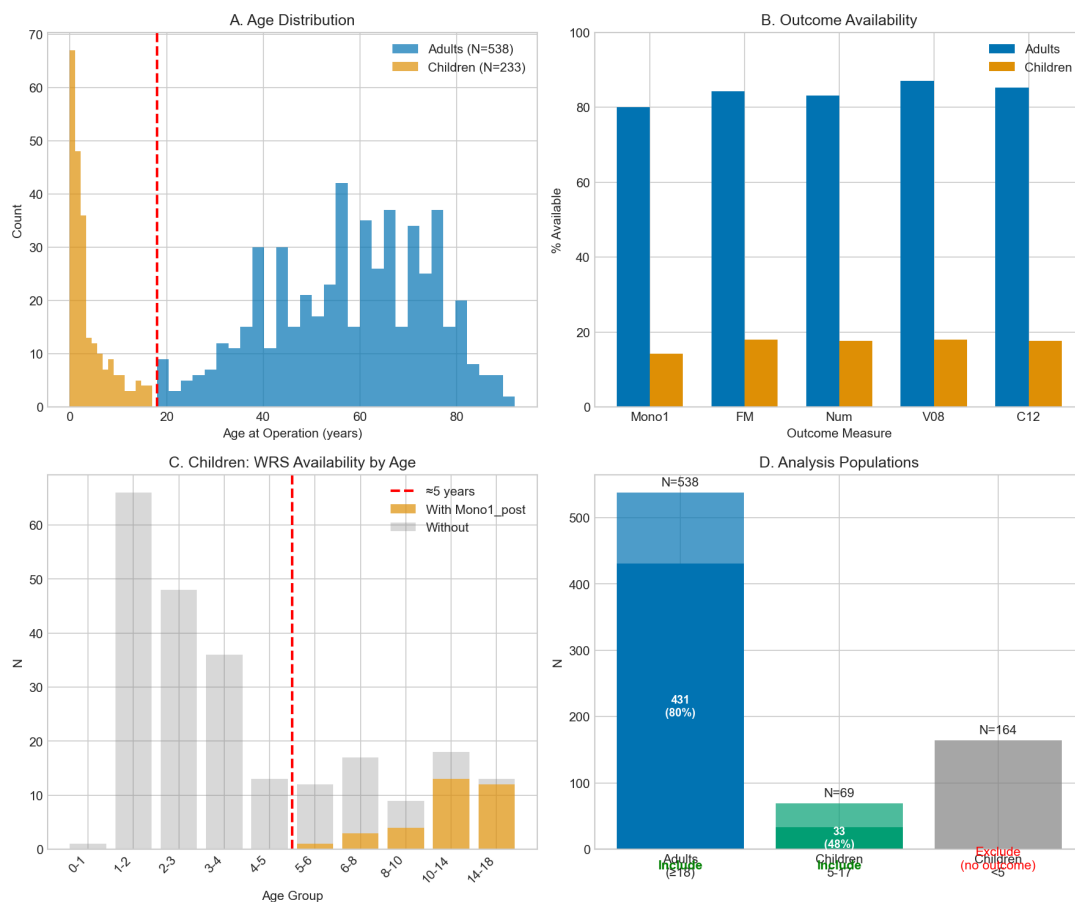


Figure 18: A: Age distribution showing bimodal pattern (pediatric peak at 1–3 years, adult peak at 50–70 years). B: Outcome availability by population. C: Children WRS availability by age (none under 5). D: Analysis populations with outcome availability.

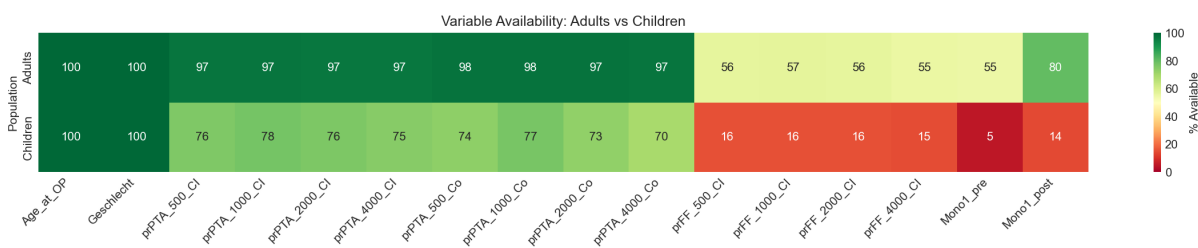


Figure 19: Variable availability comparison. Adults have higher availability for most predictors, particularly free-field audiometry and pre-operative WRS.

8.2 Outcome Engineering

Three outcome formulations were tested:

- **Continuous:** Raw WRS percentage (0–100%)
- **Binary:** Success defined as $\text{WRS} \geq 50\%$
- **Categorical:** Poor ($<55\%$), Moderate (55–80%), Good ($\geq 80\%$)

Additionally, a composite score averaging Mono1, V08, and C12 (standardized) was tested but did not improve performance.

8.3 Cross-Population Benchmark

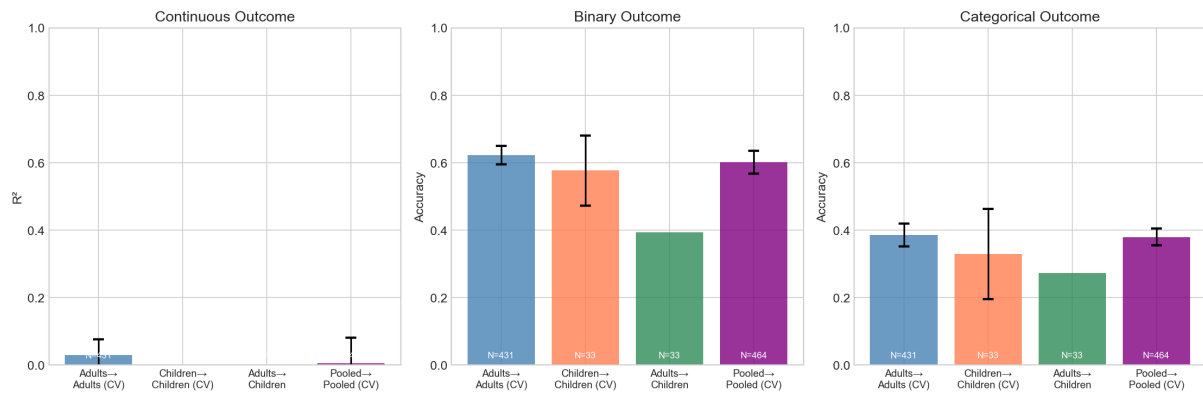


Figure 20: Cross-population benchmark results. Performance degrades substantially when adult models are applied to children.

Outcome	Train → Test	Metric	Value	N (test)
Continuous	Adults → Adults (CV)	R^2	0.03	431
	Children → Children (CV)	R^2	-3.02	33
	Adults → Children	R^2	-0.14	33
	Pooled → Pooled (CV)	R^2	0.00	464
Binary	Adults → Adults (CV)	Accuracy	0.62	431
	Children → Children (CV)	Accuracy	0.58	33
	Adults → Children	Accuracy	0.39	33
	Pooled → Pooled (CV)	Accuracy	0.60	464
Categorical	Adults → Adults (CV)	Accuracy	0.39	431
	Children → Children (CV)	Accuracy	0.33	33
	Adults → Children	Accuracy	0.27	33
	Pooled → Pooled (CV)	Accuracy	0.38	464

Table 48: Cross-population benchmark. Adult models fail to generalize to children (red values). Children-only models are unstable due to small N.

8.4 Key Findings

- Binary outcome performs best:** 62% accuracy on adults, more stable than continuous prediction
- Adult models do not transfer:** Accuracy drops from 62% to 39% when applied to children
- Children-only models are unstable:** N=33 is insufficient for reliable modeling
- Pooling does not help:** Children represent only 7% of pooled data
- Composite scores do not improve performance:** R^2 remains near zero

Implication for RQ3: Predictive factors appear to differ between adult and pediatric populations. The adult model framework (Bertschinger) does not generalize to children, justifying the need for age-stratified models or pediatric-specific predictors.

9 Pediatric Analysis: Data Gap

A separate analysis of the combined dataset (N=771 including children) identified a critical gap: children under 5 years cannot perform word recognition tests.

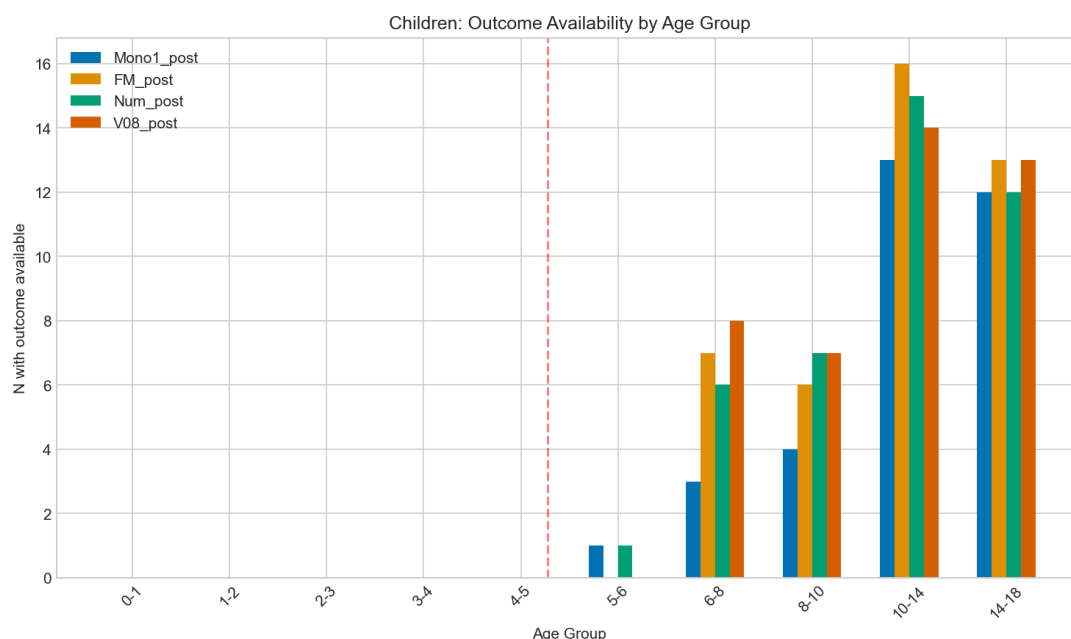


Figure 21: Children outcome availability by age group. No outcomes available below age 5; limited availability until age 10.

Age Group	N	%	WRS Available	Issue
<1 year	1	0.4%	0	Cannot perform WRS
1–2 years	66	28.3%	0	
2–3 years	48	20.6%	0	
3–5 years	49	21.0%	0	
5–10 years	38	16.3%	8	Can perform WRS
10–18 years	31	13.3%	25	Can perform WRS

Table 49: Pediatric patients (N=233). 70% are <5 years and cannot perform word recognition tests.

9.1 Data Request

To enable pediatric outcome prediction and quality-of-life analysis, we request:

1. **CAP scores** (Categories of Auditory Performance) — pre- and post-operative
2. **SIR scores** (Speech Intelligibility Rating) — pre- and post-operative
3. **LittleEars scores** — if available for infants
4. **Subjective questionnaires** (if available):
 - SSQ (Speech, Spatial, Qualities of Hearing)
 - NCIQ (Nijmegen CI Questionnaire)
 - APHAB (Abbreviated Profile of Hearing Aid Benefit)

CAP/SIR are essential for children <5 who cannot perform speech tests. Subjective questionnaires would enable quality-of-life prediction and help interpret cases where WRS declines

but patient satisfaction improves.

10 Discussion Points

10.1 Decline Cases

21.8% of patients show *decreased* WRS post-CI. Our analysis reveals:

- These are patients with **high pre-operative function** (mean 67.6% WRS)
- Post-op mean: 44.6% (vs. 70.3% for non-decline cases)
- **Not measurement noise:** Excluding them *worsens* prediction (CV R^2 drops to -0.03)
- **Interpretation:** Ceiling effect — patients with good residual hearing have limited room for improvement and risk losing acoustic hearing post-CI

10.2 Ceiling Effects

Patients with high pre-op WRS ($>70\%$) show limited improvement potential. Options:

- Stratify analysis by baseline performance
- Consider relative improvement metrics
- Accept as limitation of the outcome measure

10.3 Temporal Validation Strategy

COVID-19 dip (2020) and recent growth (2022–2024) support training on 2012–2021, testing on 2022–2024 (Figure 2). *Does this align with known protocol changes?*

10.4 Relationship to Bertschinger et al. (2024)

Our analysis replicates and extends Bertschinger’s work:

Aspect	Bertschinger	This Study
Population	Adults only	Adults + Children
N (adults)	360	431–538
Validation	Training only	Cross-validation
R^2 (adults)	0.27 (train)	0.11 (CV)
Pediatric generalization	Not tested	Fails
Outcome types	Continuous only	Continuous, Binary, Categorical

Table 50: Comparison with Bertschinger et al. (2024).

11 Next Steps

11.1 Completed

1. **Bertschinger replication:** CV $R^2=0.11$ (vs. her reported 0.27 training)
2. **ML benchmark:** XGBoost, RF, LightGBM — all underperform linear model
3. **Temporal validation:** Ridge $R^2=0.07$ on 2022+ data
4. **Cross-population:** Adult model fails on children
5. **Uncertainty quantification:** 90% intervals span $\pm 39\%$ WRS
6. **Conformalized Quantile Regression:** Adaptive intervals (46–112% range)
7. **Sensitivity analyses:** Interactions, transformations, subgroups — no improvement
8. **Bootstrap CI:** CV R^2 95% CI = $[0.044, 0.210]$

9. **Extended predictor analysis:** All 226 variables, Multi-Task Lasso CV $R^2=0.18$
10. **Simpson's Paradox analysis:** Overall $r=0.16$ is between-group artifact; within-stratum $r \approx 0$
11. **Latent factor models:** Multi-output, factor analysis, interactions — pre/post measure different systems
12. **Decline prediction:** 24% decline cases, CV AUC=0.83, high baseline = high risk
13. **Clinical counseling guide:** Baseline-stratified outcome expectations and decline risk

11.2 Remaining

1. **External validation:** Hannover data (if available)
2. **Pediatric analysis:** Awaiting CAP/SIR data
3. **Draft manuscript**

11.3 Pending Data

1. CAP/SIR scores for children <5
2. Subjective questionnaires (SSQ, NCIQ) if available

12 Timeline

Milestone	Target
Baseline replication	Complete
Adult vs. pediatric comparison	Complete
ML benchmark (XGBoost, RF, LightGBM)	Complete
Extended predictor analysis	Complete
Temporal validation	Complete
Uncertainty quantification	Complete
Conformalized Quantile Regression	Complete
Sensitivity analyses (interactions, transforms)	Complete
Bootstrap confidence intervals	Complete
Phenotype discovery (clustering)	Complete
Simpson's Paradox & pre-post correlation	Complete
Latent factor & multi-output models	Complete
Decline prediction analysis	Complete
Clinical counseling guide	Complete
Pediatric data integration (CAP/SIR)	Upon receipt
Draft manuscript	February 2026

13 Notebooks & Outputs

All analysis code is in `CI_Outcome_Pred/code/notebooks/`:

Notebook	Purpose
00_data_comparison	Adult vs. pediatric dataset comparison
01_eda	Exploratory data analysis
01_outcome_engineering	Outcome formulations & cross-population benchmark
02_baseline_bertschinger	Full Bertschinger replication (24 features, ABESS)
02b_ssd_diagnostic	SSD classification analysis
03_imputation_comparison	Compare 5 imputation methods
04_response_analysis	Test outcome transformations
04b_abess_selection	ABESS feature selection
05_ml_benchmark	ML models vs. linear baseline
06_uncertainty_quantification	Conformal prediction intervals
07_rq1_clustering	Phenotype discovery (K-means, GMM, Hierarchical)
08_temporal_analysis	Temporal trends & validation (train ≤ 2021 , test ≥ 2022)
09_extended_feature_analysis	49 available + 12 engineered features, Lasso selection
10_alternative_approaches	Imputation, model complexity, binary classification
11_prediction_improvement	Interactions, transforms, subgroups, CQR, bootstrap CI
12_correlation_cca_pls_analysis	Correlation structure, CCA, PLS, Multi-Task Lasso
13_latent_factor_models	Multi-output models, factor analysis, two-stage models
14_prepost_simpson_paradox	Simpson's Paradox, Equalizer effect, decline analysis, clinical guide

Table 51: Analysis notebooks. All code is in `CI_Outcome_Pred/code/notebooks/`.

14 Action Items

For USZ team:

1. Confirm availability of CAP/SIR data for pediatric patients
2. Review cross-population benchmark findings (adult models fail on children)
3. Confirm interpretation of decline cases (ceiling effect, not noise)
4. Review clinical counseling guide for accuracy and clinical utility

Completed by Marta:

- ✓ ML pipeline (XGBoost, RF, LightGBM) — linear model outperforms all
- ✓ Temporal validation — Ridge $R^2=0.07$ on 2022+ data
- ✓ Uncertainty quantification — 90% intervals span $\pm 39\%$ WRS
- ✓ Conformalized Quantile Regression — adaptive intervals (46–112%)
- ✓ Sensitivity analyses — interactions, transforms, subgroups tested
- ✓ Bootstrap CI — CV R^2 95% CI = [0.044, 0.210]
- ✓ Phenotype discovery — 3 clusters identified ($p=0.0004$)
- ✓ Extended predictor analysis — Multi-Task Lasso CV $R^2=0.18$
- ✓ Simpson's Paradox — within-stratum $r \approx 0$, overall $r=0.16$ is artifact
- ✓ Latent factor models — pre/post don't share latent structure
- ✓ Decline analysis — 24% decline, AUC=0.83, high baseline = high risk
- ✓ Clinical counseling guide — baseline-stratified expectations

Pending:

1. Pediatric analysis framework (awaiting CAP/SIR data)
2. External validation (Hannover data, if available)
3. Draft manuscript

— *End of Report* —